

ORIGINAL ARTICLE

A Demand Theory of the Price Level

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ABSTRACT

Heterogeneous agent incomplete markets models offer a new perspective on price and inflation determination. In contrast to complete markets, the price level is determined from the asset-market clearing condition. Fiscal and monetary policy then jointly and uniquely determine the finite steady-state price level and the inflation rate, including in a steady state in which the nominal interest rate is constant. Fiscal policy can determine the long-run inflation rate for a fiscal rule which sets the growth rate of nominal government debt, whereas both fiscal and monetary policy determine the long-run inflation rate under different tax rules.

JEL Classification: D52, E31, E43, E52, E62, E63

1 | Introduction

The prevailing view on monetary policy (Woodford 2003; Galí 2015) is that it works through setting nominal interest rates, that it controls the inflation rate, and that prices are determinate if monetary policy responds sufficiently strongly to inflation—the Taylor principle. Fiscal policy is largely irrelevant in this view.

In this paper, I propose that heterogeneous agent incomplete markets models offer a new and different perspective on these issues. While monetary policy continues to control the nominal interest rate, fiscal policy is now assigned a significant role. Two results stand out.

First, in contrast to the conventional view, in the absence of output growth, and even when monetary policy operates an interest rate rule with a different inflation target, the long-run inflation rate is equal to the growth rate of nominal fiscal variables, which are controlled by fiscal policy. In this scenario, a

tough, independent central bank is not only insufficient to ensure price stability in the long-run, it also has no direct control over long-run inflation, even if it follows an interest rate rule which satisfies the Taylor principle.¹ The fiscal determination of long-run inflation leads to a reinterpretation of one of Friedman's (1963, 1968, 1970) key propositions that “inflation is always and everywhere a monetary phenomenon in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output.” Friedman assumed that money supply, purposefully and astutely controlled by central banks, serves as a nominal anchor to control inflation. Fifty years after Friedman (1963;1968;1970) however, central bank practice and academic research has shifted to using a short-term nominal interest rate as the policy instrument. The money supply is no longer controlled by monetary policy, but determined endogenously as clearing the money market. I incorporate these features into the model, which strips money of its anchoring role envisaged by Friedman. However, the nominal fiscal variables, controlled by fiscal policy, such as nominal government debt and spending, replace money in this role.

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Second, incomplete markets models provide a new theory of price-level determination, in which the price level is determined as clearing the asset, or equivalently the goods market. Monetary policy works through setting nominal interest rates, and fiscal policy sets nominal government spending, taxes, and government debt, satisfying the present-value government budget constraint at all times and for all prices. That is, the fiscal policy is passive (Leeper 1991). In this environment, I show that the steady-state price level is determinate, even if nominal interest rates are constant. The price level is determined jointly by monetary and fiscal policy, and the long-run inflation rate by fiscal policy only.

To understand these results, it is sufficient to combine a few simple insights. First, market incompleteness generates a precautionary savings demand, such that the household sector effectively values real government bonds, that is, their real value exceeds the value of tax liabilities. When, in addition, government bonds are nominal, shifts in the price level shift the real value of debt. The price level is then determined using the asset-market clearing condition, such that the demand for real bonds equals its supply. In contrast, if markets were complete, Ricardian equivalence would hold, and the household sector would be indifferent to the amount of government bonds available, leaving the price level indeterminate.

Second, the steady-state real interest rate is not determined uniquely by a household discount factor, but instead depends on the amount of real bonds available. In incomplete markets models, households are willing to accept a lower return than suggested by their discount factor, as bonds allow households to smooth consumption more effectively in response to uninsurable idiosyncratic income shocks (Aiyagari 1995). Bonds have a “liquidity premium,” which, under standard assumptions, is decreasing in the amount of bonds households own. To absorb more government bonds, households thus require a higher real interest rate. This indicates that, depending on the amount of bonds available, a continuum of steady-state real interest rates is feasible. Monetary and fiscal policy now jointly choose one out of this continuum of potential steady-state real interest rates. Monetary policy sets the steady-state nominal interest rate, whereas fiscal policy sets the growth rate of nominal government debt. In a steady state, the value of real government debt is constant, such that the steady-state condition for fiscal policy is that the growth rate of nominal debt equals the inflation rate (in the absence of economic growth). The real interest rate is then determined by the Fisher equation, as the ratio of the nominal interest rate to the inflation rate. Clearly, this logic for determining the long-run inflation rate does not apply if the steady-state real interest rate is pinned down by the discount factor, as is the case in standard New Keynesian complete markets models.

Section 3 builds on these simple insights and shows that the steady-state price level is determinate in a large class of heterogeneous agent incomplete markets models. I also explain the mechanism behind the determinacy result; why incomplete markets models deliver determinacy and why complete markets models lead to indeterminacy when nominal interest rates are constant; why fiscal policy has to be partially nominal; why the Demand Theory of the Price Level (DTPL) proposed here is not the Fiscal Theory of the Price Level (FTPL); and why adding capital to the model does not alter these conclusions.

This paper offers a new solution to the Sargent and Wallace (1975) challenge of obtaining price-level determinacy when the nominal return on bonds is pegged. These authors considered the classic policy problem of a central bank which can either set a constant nominal return on government bonds or set money supply, but not both independently at the same time. To capture this environment, I enrich the model to a two-asset economy in which money and bonds coexist and can earn different interest rates. Again, the steady-state price level is determinate when monetary policy sets the nominal interest rate on bonds and does not set money supply, which becomes an endogenous variable.

I then consider a generalization with nominal government expenditure, in order to illustrate the role of aggregate demand in my theory. To understand the role of the government in managing aggregate nominal demand, I build on the comparative static results for steady states presented in Acemoglu and Jensen (2015). Taking prices as given, I prove that a higher steady-state lump-sum tax rebate increases household steady-state consumption less than one-for-one and thus also increases savings. This result implies that taking prices as given, lower steady-state government spending by one dollar and a simultaneous higher tax rebate of the same amount to private households lowers real total aggregate demand—the sum of private and government demand. The same logic also establishes why in a steady state, real aggregate demand depends on the price level. Given monetary and fiscal policy, a higher steady-state price level lowers real government consumption, since government spending is (partially) fixed in nominal terms. At the same time, it lowers the tax burden for the private sector by the same amount. As private sector demand does not substitute one-for-one for the drop in government consumption, but instead saves a fraction of the tax reduction, aggregate real demand falls, establishing a downward-sloping aggregate demand-price curve. The unique equilibrium steady-state price level is then such that aggregate real demand equals real supply. In particular, the price level is also determinate if bonds are fully real or even if bonds are in zero net supply, explaining why it is more generally a “A Demand Theory of the Price Level” and not just a “demand for bonds” theory. The price level and the inflation rate are ultimately determined by the level and the growth rate of nominal aggregate demand, which in turn is derived from policy choices of nominal government bonds, nominal government expenditures, nominal taxes and transfers, and nominal interest rates.²

My theory does not rely on fiscal policy following simple rules such as setting a constant debt growth rate. Instead, the price level is determined as clearing the asset market and the growth rate of nominal debt is merely a sufficient statistic for fiscal policy.³ Based on this insight, I establish how the price level is determined for various tax rules, including active and passive policies as in Leeper (1991), which do not set the growth rate of nominal debt. I show that the finite steady-state price level is uniquely determined if the tax rule implies a unique growth rate of nominal government debt. This is the case if the tax rule is specified in nominal terms but not necessarily if written in real terms. The long-run inflation rate is again equal to the steady-state growth rate of nominal debt, which can now depend on the conduct of both fiscal and monetary policy.

I show that active and passive fiscal rules imply the same steady-state price level and inflation rate if they induce the same growth rate of nominal debt. In particular, active and passive fiscal policies feature the same determinacy properties in contrast to complete markets models (Leeper 1991). Building on a large literature on the FTPL in complete markets models, Brunnermeier et al. (2023) and Kaplan et al. (2023) incorporate the FTPL into incomplete markets models. I show that the steady-state price level is determined according to the DTPL in both papers. I also explain why the authors' FTPL interpretation of their results does not fully reflect the economic mechanisms operating in their models. Recently, Angeletos et al. (2024) also use the DTPL developed in this paper to study how deficits induce inflation.

It is important to emphasize that these results do not hold in all models in which Ricardian equivalence fails. For example, the price level is not determinate in an economy where a fraction of households simply consume their current income “hand-to-mouth,” while the remaining households act according to the permanent income hypothesis (PIH).⁴ The reason for the indeterminacy is that only permanent-income households hold bonds, and thus, shifts in the value of public debt have no aggregate demand effects, but only shift consumption from one group to the other. Similar arguments apply to the perpetual youth model and its variants (Yaari 1965; Blanchard 1985; Bénassy 2005;2008), in which Ricardian equivalence does not hold, but the price level is indeterminate, as explained in the Appendix.

This paper shows that using the workhorse incomplete markets model not only provides an empirically superior model of consumption (Kaplan and Violante 2014)—a key part of the monetary transmission mechanism—but also entails a windfall gain: the price level is determinate. This seems important, since a growing body of literature has recently emerged which incorporates price rigidities into heterogeneous agent incomplete markets models (HANK [Heterogeneous Agents New Keynesian]).⁵ One motivation to do so is that, while able to generate a realistic distribution of marginal propensities to consume, the textbook incomplete markets model does not allow output to be demand-determined, as prices are fully flexible, potentially limiting its applicability to many questions raised by the Great Recession. Adding a nominal side to the model and allowing for price rigidities however, forces us to address the same questions we confront in complete markets models: How is the price level determined? What type of monetary and fiscal policies ensure determinacy? How does taking the zero lower bound (ZLB) into account affect the conclusions derived from the model? The determinacy result in this paper provides the foundations for obtaining these answers. The results are derived within the standard framework in macroeconomics and dynamic public finance—heterogeneous agent incomplete markets models—and are obtained without invoking any policy assumptions except that it is partially nominal. Yet, the two facts—markets are incomplete and government bonds are nominal—inevitably implies a new theory of price-level determinacy, namely, DTPL. Section 4 concludes and considers four contributions in particular, Hagedorn (2016) and Hagedorn et al. (2017;2024b;2019), who use incomplete markets models which build on the answers and insights provided in this paper so as to consider forward guidance, the fiscal multiplier and several

other puzzles which arise in New Keynesian models during a liquidity trap.

2 | A Heterogeneous Agent Incomplete Markets Model

In this section, I describe a heterogeneous agents endowment economy with uninsurable idiosyncratic labor income risk in which markets are incomplete, based on Huggett (1993), where only a government bond can be traded, subject to exogenously imposed borrowing limits. I consider a cashless economy as in Woodford (2003) where monetary policy sets the nominal interest rate but allow for a nontrivial demand for money in Section 3.7. The household sector is fully described in real terms, as this allows for a clean definition of aggregate asset demand in real terms in Section 2.4.

2.1 | Households

The economy is populated by a continuum of households of measure one with preferences over consumption c_t

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \quad (1)$$

where u is increasing and strictly concave, β is the subjective discount factor and the expectation is taken over realizations of idiosyncratic endowments. Agent endowment $\{e_t\}_{t=0}^{\infty}$ is stochastic and characterized by an N -state Markov chain that can take on values $e_t \in \mathcal{E} = \{e_1, \dots, e_N\}$ with the transition probability characterized by $\vartheta(e'|e)$ and $\int e = 1$. A household saves real bonds a_{t+1} in Period t subject to the credit constraint

$$a_{t+1} \geq -\bar{a} \quad (2)$$

for a credit limit $\bar{a}$.⁶ The real return is $1 + r_{t+1}$ and thus, at time t , the household faces the budget constraint

$$c_t + a_{t+1} = (1 + r_t)a_t + e_t - \tau_t, \quad (3)$$

where τ_t is a lump-sum tax. The household choice problem can be written recursively, given initial conditions (a_0, e_0) ,

$$V_t(a, e; \Omega) = \max_{c \geq 0, a' \geq -\bar{a}} u(c) + \beta \sum_{e' \in \mathcal{E}} \vartheta(e'|e) V_{t+1}(a', e'; \Omega') \quad (4)$$

$$\text{subject to } c + a' = (1 + r_t)a + e - \tau_t$$

$$\Omega' = \mathcal{T}_t(\Omega),$$

where a prime denotes the next period's value, $\Omega(a, e) \in \mathcal{M}$ is the distribution on the space $X = \mathcal{A} \times \mathcal{E}$, agent asset holdings $a \in \mathcal{A}$, and endowment $e \in \mathcal{E}$, across the population, which, together with the policy variables, determine the equilibrium prices. $\mathcal{T}$ is an equilibrium object that specifies the evolution of the distribution Ω in the set of probability measures $\mathcal{M}$ over X with σ -algebra $\mathbb{B}(X)$. The transversality condition is

$$\lim_{T \rightarrow \infty} E_0[\beta^T u'(c_T) a_{T+1}] = 0. \quad (5)$$

2.2 | Government

Monetary policy sets the nominal interest i_t , the aggregate Period t price level is P_t , the real interest rate $1 + r_{t+1} = (1 + i_{t+1})P_t/P_{t+1}$ and the inflation rate is $1 + \pi_{t+1} = P_{t+1}/P_t$.⁷ Since the steady-state real interest rate can be negative and there is yet no money in this model, negative nominal interest rates are possible. Fiscal policy sets the nominal level of bonds B_t and adjusts nominal lump-sum taxes T_t to satisfy the government budget constraint at all times⁸

$$T_t := (1 + i_t)B_t - B_{t+1}, \quad (6)$$

so that households pay taxes $\tau_t = T_t/P_t$ in equilibrium.

2.3 | Equilibrium

The asset-market clearing condition is

$$\frac{B_{t+1}}{P_t} = \int a_{t+1} d\Omega_t \quad (7)$$

and the resource constraint reads

$$\int c_t d\Omega_t = \int e_t d\Omega_t = 1. \quad (8)$$

A *competitive equilibrium* is then a sequence of prices P_t , i_t , r_t , taxes T_t , bonds B_t , value function V_t with policy functions c , a' , and a law of motion $\mathcal{T}_t : \mathcal{M} \rightarrow \mathcal{M}$ such that:

1. Households maximize utility, taking prices and government policies as given.
2. The government budget constraint is satisfied.
3. The resource constraint is satisfied.
4. The transversality condition (5) holds.
5. The asset market clears.
6. The aggregate law of motion $\mathcal{T}_t$ is generated by a' and ϑ .

A *stationary competitive equilibrium* is a competitive equilibrium where Ω_t , $1 + r_t$, $1 + \pi_t$, and $1 + i_t$ are constant and

$$\frac{B_t}{B_{t-1}} = \frac{T_t}{T_{t-1}} = \frac{P_t}{P_{t-1}} = 1 + \pi_{ss}. \quad (9)$$

Since this paper is not about hyperinflations and hyperdeflations, I only consider finite nonzero price levels.

2.4 | The Asset Demand Function

It is standard to characterize the stationary equilibrium through the asset-market clearing condition and I follow this approach here. Given initial period t assets a_t , sequences of real interest rates $(r_t, r_{t+1}, r_{t+2}, \dots)$, and taxes $(\tau_t, \tau_{t+1}, \dots)$, each household maximizes utility yielding a sequence of assets $(a_{t+1}, a_{t+2}, \dots)$. Aggregate asset demand at time t ,

$$S_{t+1} = \int a_{t+1} d\Omega_t, \quad (10)$$

is then a function of the initial distribution of assets Ω_t , real interest rates, and taxes,

$$S_{t+1} = S(\Omega_t; 1 + r_t, 1 + r_{t+1}, 1 + r_{t+2}, \dots; \tau_t, \tau_{t+1}, \dots). \quad (11)$$

In a stationary equilibrium, the real interest rate r_{ss} , aggregate asset demand S_{ss} , and the distribution of assets Ω_{ss} are invariant, and taxes $\tau_{ss} = r_{ss}S_{ss}$. Steady-state asset demand is the fixed point of

$$S_{ss} = S(\Omega_{ss}; 1 + r_{ss}, 1 + r_{ss}, 1 + r_{ss}, \dots; r_{ss}S_{ss}, r_{ss}S_{ss}, \dots), \quad (12)$$

such that steady-state asset demand is a function $S(1 + r)$ of the real interest rate only.⁹ The equilibrium is characterized through the asset-market clearing condition

$$S(1 + r) = \frac{B}{P}, \quad (13)$$

since equilibrium conditions 1, 2, and 4 are satisfied by construction of the S function, and the asset-market clearing (condition 5) implies that the resource constraint (condition 3) is satisfied.¹⁰ If the inflation rate π is not zero, bonds in a steady state issued at time t equal $B(1 + \pi)^t$ and the price level equals $P(1 + \pi)^t$ for initial values B and P , so that the term $(1 + \pi)^t$ term cancels itself out when computing the real value of bonds B/P .

3 | Steady-State Price-Level Determinacy

In this section, I establish that the steady-state price level is determinate. This is particularly difficult, since the nominal interest rate on bonds is constant in a steady state and therefore, the Sargent and Wallace (1975) critique applies fully. One way to understand why the price level in complete markets models is indeterminate is to note that the number of endogenous variables exceeds the number of equilibrium conditions by one. I show that the asset-market clearing condition in incomplete markets models provides an additional equation that is needed to determine the price level. The formal analysis is complemented with a graphical one, which together show that this result holds not only in a large class of incomplete markets models, but in all models with a steady-state asset demand curve, which describes a trade-off between the real return and the quantity of the asset. Prices are assumed to be flexible throughout, since price stickiness is irrelevant for steady-state determinacy, simply because a steady state requires no price adjustment other than an increase at the steady-state inflation rate (Nakajima and Polemarchakis 2005).¹¹ To convey the intuition more effectively, I start with a special case of my determinacy result in a model which resembles the well-understood Bewley monetary economy as much as possible. I then generalize the result to models which cannot be mapped into the Bewley economy. In one model, government expenditure is nominal but bonds are real and in the other one money and bonds coexist. These generalizations also make clear why it is a “Demand theory of the price level” and not just a “demand for bonds theory” since aggregate nominal (government) demand and not the amount of outstanding government bonds ultimately determines prices and inflation.

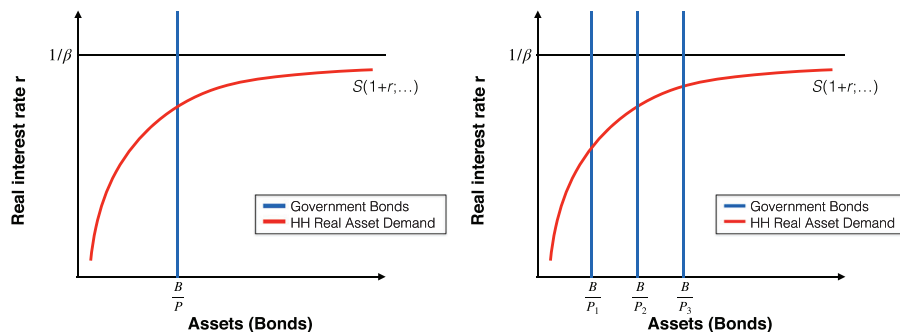


FIGURE 1 | Asset market in incomplete markets economy.

3.1 | Asset-Market Clearing and Price-Level Determinacy

As is well-known, the incomplete markets economy is in equilibrium if aggregate asset demand (households' assets) equals real aggregate asset supply (government bonds), which can be represented by the well-known Figure 1 (left panel).¹² Household asset demand $S(1+r)$ is an upward-sloping function of the real interest rate $1+r$ which is smaller than $1/\beta$. Real steady-state asset supply from the government equals

$$\frac{B}{P} = \frac{B(1+\pi)^t}{P(1+\pi)^t} = \frac{B_{t+1}}{P_t},$$

such that the equilibrium condition is

$$S(1+r) = \frac{B}{P}. \quad (14)$$

This is one equation with two unknowns, the real interest rate $1+r$ and the price level P . This suggests that a continuum of price levels (associated with a continuum of real interest rates), for example, P_1, P_2, P_3 , clears the asset market as illustrated in the right panel of Figure 1. Before establishing that Equation (14) nevertheless determines the price level, since the real interest rate is determined by monetary and fiscal policy, I discuss the Bewley (1980) monetary model to clarify the workings and the conceptual contribution of my theory.

The specification of fiscal policy is so far intentionally simple. For example, government expenditure is absent and government bonds is the only asset as households can neither hold money nor physical capital. The benefit of the simplicity is that it renders my analysis comparable to the Bewley (1980) basic model of fiat money with a well-understood intuition described in textbooks.

To explain the mapping between the Bewley economy and this special case of my theory, I follow the exposition of the Bewley model in Ljungqvist and Sargent (2012) but use my notation. The central bank provides a fixed supply of money M , the only asset in this economy. The derivation of the aggregate money demand function in Bewley, $L(\pi_{ss})$, is basically identical to the derivation of my S function. The steady-state inflation rate is zero so that the real return on money is 0. The steady-state price level P^* is then determined as a solution to the money-market clearing condition

$$\frac{M}{P^*} = L(0), \quad (15)$$

which looks like replacing M with B and L with S in (14). If the central bank pays an interest rate i^m on money holdings, the money-market clearing condition

$$\frac{M}{P^*} = L(1+i^m) \quad (16)$$

is formally equivalent to (14) and the Bewley intuition carries over to my model.

The formal equivalence, however, masks the conceptual differences between the two approaches, which are important to understand the Sargent and Wallace (1975) critique and how it is overcome in this paper. Sargent and Wallace (1975) argue that a monetary authority setting the nominal return on bonds and not setting money supply, as in my model, induces price-level indeterminacy. Instead setting money supply exogenously, as in the Bewley model, implies determinacy.

This paper successfully addresses this challenge of delivering price-level determinacy when monetary policy sets the nominal return on bonds. Although the central bank also sets an interest rate when paying interest on money this is not what the Sargent and Wallace (1975) critique is about. The idea is rather that the central bank can either set money supply or the nominal return on bonds but not both independently at the same time. While the above equivalence might blur this conceptual distinction, the difference will be clear in models in which money and bonds are different assets which can earn different interest rates (as observed in the data). Section 3.7 therefore considers this two-asset model, which addresses the Sargent and Wallace (1975) critique, delivers price-level determinacy, but cannot be mapped into the Bewley model. As a further generalization, Section 3.8 establishes price-level determinacy if bonds are real and government expenditure is nominal. Before turning to these generalizations, I will first explain the determinacy result in the simple case with nominal bonds but without money or government expenditure.

3.2 | How Monetary and Fiscal Policy Determine the Steady-State Real Interest Rate

A key step in the argument is to show how monetary and fiscal policy determine the steady-state real interest rate. In both complete and incomplete markets models, a Fisher relationship between the steady-state nominal interest i_{ss} , real interest rate r_{ss} , and inflation π_{ss} holds:

$$1 + r_{ss} = \frac{1 + i_{ss}}{1 + \pi_{ss}}. \quad (17)$$

Monetary policy sets the steady-state nominal interest rate i_{ss} . Fiscal policy sets the growth rate of nominal debt (B) and adjusts nominal tax revenue (T) to balance the government budget. In a steady state, real tax revenue and real government debt are constant, such that the steady-state condition for fiscal policy is that the growth rates of nominal tax revenue and nominal debt all equal the inflation rate in the absence of economic growth (a prime denotes the next period's value),¹³

$$\pi_{ss} = \frac{B' - B}{B} = \frac{T' - T}{T}. \quad (18)$$

Nonconstant growth rate policies would be inconsistent with a steady state and are considered in the local determinacy analysis in Hagedorn (2023). Note that a specific interpretation is assigned to these steady-state conditions: If fiscal policy decides on a 2% nominal growth rate in nominal debt, $(B' - B)/B$, then the steady-state condition for steady-state real government debt to be constant requires the steady-state inflation rate to equal 2% as well. The steady state further requires nominal tax revenue T also to grow at 2%. It is important to note that these considerations do not determine the levels of real taxes and real debt, except in the sense that these are unchanging over time in a steady state. In particular, the price level has not yet been determined.

Equation (18) means that the inflation rate is equal to the growth rate of nominal government debt, implying that the equilibrium real interest rate is determined jointly by monetary and fiscal policy. Monetary and fiscal policy cannot simply implement an arbitrary steady-state real interest rate. Restricting the real interest rate $1 + r_{ss}$ such that $\beta(1 + r_{ss}) < 1$ and $S(1 + r_{ss}) > 0$ ensures that a steady state with a finite nonzero price level exists. If $\beta(1 + r_{ss}) \geq 1$ asset demand would become infinite, a well-known result in incomplete markets models, and no steady state with a $P \in \mathbb{R}_{>0}$ exists. Similarly, if $S(1 + r_{ss}) \leq 0$, no steady state $P \in \mathbb{R}_{>0}$ would clear the asset market.

These conclusions about the steady-state inflation rate are valid, even if monetary policy implements an interest rate rule such as

$$i_{t+1} = \max(\bar{i} + \phi(\pi_t - \pi^*), 0) \quad (19)$$

for an inflation target π^* , an intercept $\bar{i}$, and $\phi > 0$. In this case, inflation is still determined by Equation (18) and the steady-state nominal interest rate equals¹⁴

$$i^{ss} = \max(\bar{i} + \phi\left(\frac{B' - B}{B} - \pi^*\right), 0). \quad (20)$$

Note that this line of reasoning requires a continuum of potential steady-state real interest rates, and not just one equal to $1/\beta$, as in complete markets models. Therefore, this logic for determining the long-run inflation rate does not apply if markets are complete.

3.3 | Price-Level Determinacy

I can now use Equation (14) to determine the unique finite steady-state price level. Using the result that $1 + r_{ss} = (1 + i_{ss})/(1 + \pi_{ss})$ is set by policy to eliminate the real interest rate from the list of unknowns, Equation (14) now has just one unknown, the price level P^* :

$$S\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}\right) = \frac{B}{P^*}, \quad (21)$$

which serves to determine the unique steady-state price level, as illustrated in the left panel of Figure 2, simply as

$$P^* = \frac{B}{S\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}\right)} \in \mathbb{R}_{>0} \quad (22)$$

since $\beta(1 + r_{ss}) < 1$, $S(1 + r_{ss}) > 0$ and $B > 0$. Note that here uniqueness is obtained since the real interest rate is determined by monetary and fiscal policy. For every real interest rate, the stationary distribution and aggregate asset demand are uniquely determined under standard assumptions (e.g., Hopenhayn and Prescott 1992; Açıkgöz 2018) and a unique price level then ensures asset market clearing.¹⁵ In particular, the asset demand curve does not have to be upward sloping and the monotonicity in Figure 2 is just for illustrative purposes.¹⁶ Uniqueness of the stationary distribution just ensures that aggregate asset demand is a function and not a correspondence and does not restrict the slope of the asset demand function. And asset demand being a function—a standard assumption—is all what is needed for my results to hold.

There are two key assumptions to obtain price-level determinacy. First, fiscal policy is nominal. Without this assumption, fiscal policy would be specified fully in real terms, and the equilibrium in the asset market would not depend on the price level, such that the equilibrium condition cannot be used to determine the price level. The price level is however determinate in the presence of a debt/GDP target. This is the case simply because Period t nominal debt is issued in Period $t - 1$ and is thus a predetermined nominal variable in Period t and serves as an anchor for the Period t price level. Targeting next period's debt/GDP ratio then requires understanding how this period nominal debt level and the next period price level are connected in equilibrium. The government then has to issue the right amount of nominal debt B_{t+1} today such that the price level P_{t+1} (determined tomorrow) delivers the desired debt/GDP ratio, $B_{t+1}/(P_{t+1}Y_{t+1})$, tomorrow.¹⁷ Second, there is a steady-state aggregate asset demand function, which depends on the real interest rate. This is a standard result in models with heterogeneous agents and market incompleteness. I explain below why the arguments supporting the incomplete markets economy do not apply in complete markets environments.

The infinite horizon assumption ensures a nondegenerated asset demand curve in each period and in particular in a steady state that can be used to determine the price level. To understand this

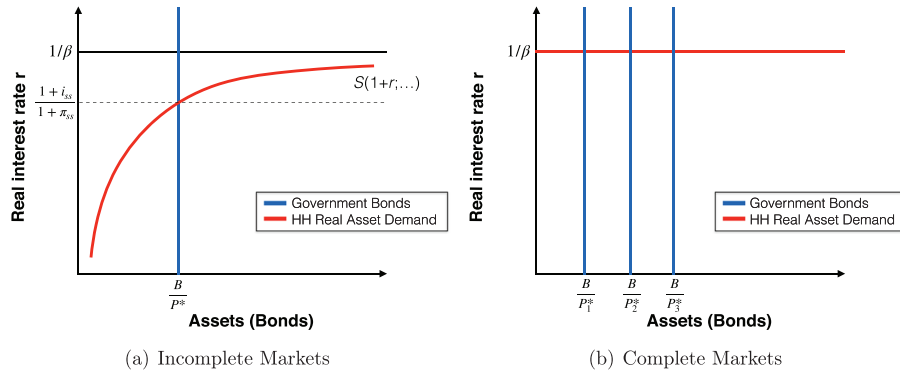


FIGURE 2 | Asset market equilibrium: (a) Determinacy and (b) Indeterminacy.

better, assume a finite horizon such that in the last period, the demand for bonds is zero. Asset market clearing then cannot be used to determine the price level, rendering the price level indeterminate. This last-period indeterminacy carries over to previous periods, such that the price level would be indeterminate in all periods, as, for example, in Geanakoplos and Mas-Colell (1989) and Balasko and Cass (1989).

3.4 | Representative Agent and Hand-to-Mouth Consumers

The above reasoning does not extend to representative agent environments, so that the price level is indeterminate if markets are complete. The key implication of complete markets is that the steady-state real interest rate is determined by the discount factor only, $(1 + r_{ss})\beta = 1$, whereas in incomplete markets models, the real interest rate depends on virtually all model primitives, so that the equilibrium in the asset market can be represented equivalently as

$$\frac{1 + i_{ss}}{1 + \pi_{ss}} = 1 + r_{ss}(B/P^*), \quad (23)$$

which again determines the steady-state price level P^* and where $r_{ss}(B/P)$ is the real interest rate which makes households willing to hold B/P real assets in a steady state. The counterpart in a representative agent model is

$$\frac{1 + i_{ss}}{1 + \pi_{ss}} = 1 + r_{ss} = 1/\beta, \quad (24)$$

which no longer depends on the price level, which is therefore indeterminate. The right panel of Figure 2 illustrates the indeterminacy, depicting supply and demand in the asset market as before, with incomplete markets. The difference is that the steady-state asset demand curve is a vertical line at the steady-state interest rate $1/\beta$, whereas it is an upward-sloping curve in the model with incomplete markets. An equivalent interpretation of the reason for indeterminacy is that the representative household is willing to absorb any amount of real bonds in a steady state with real interest rate $1/\beta$, whereas asset demand is unique for every feasible real interest rate if markets are incomplete. The vertical asset demand curve with complete markets then reflects the result that the real interest rate is independent of the quantity

of real bonds, such that a continuum of price levels, for example, P_1^*, P_2^*, P_3^* , satisfies all equilibrium conditions.

The same arguments apply to models in which a fraction of households simply consume their current income, “hand-to-mouth,” while the remaining households act according to the PIH. Since hand-to-mouth consumers do not participate in the asset market, the real interest rate is determined by the discount factor of PIH households only, $(1 + r_{ss})\beta = 1$, and equilibrium in the asset market is again characterized through

$$\frac{1 + i_{ss}}{1 + \pi_{ss}} = 1 + r_{ss} = 1/\beta, \quad (25)$$

which does not depend on the price level, implying that it is indeterminate. This model shows that it is not heterogeneity by itself that leads to the result. Rather it is the combination of heterogeneity and market incompleteness that leads to precautionary savings and a nondegenerated aggregate asset demand function, implying price-level determinacy. By the same argument, permanent heterogeneity in productivity, but otherwise complete markets, will not lead to price-level determinacy either, since again $(1 + r_{ss})\beta = 1$ in a steady state.

The graphical representation also suggests another simple way to understand why the price level is indeterminate in complete markets, but determinate in incomplete markets models. The number of endogenous variables exceeds the number of equilibrium conditions by one if markets are complete, rendering the price level indeterminate. Market incompleteness provides an extra nonredundant equation, namely, the asset-market clearing condition (21). At first glance, the argument might seem wrong, since the asset market has to clear in both models. That is correct, but in the first model, this equation is redundant, as it merely ensures that the steady-state real interest rate is equal to $1/\beta$, with no role whatsoever for bonds or the price level. In contrast, the asset demand curve in incomplete markets models defines a trade-off between the real interest rate and the value of bonds. This extra and nonredundant clearing condition, together with $1 + r_{ss} = (1 + i_{ss})/(1 + \pi_{ss})$, then determines the price level as illustrated analytically and graphically above. Appendix A shows that the reasoning for incomplete markets does not extend to perpetual youth models or to representative agent models with aggregate risk, so that the price level is also indeterminate in these models.

3.5 | Nominal and Real Tax Rules: Nominal Debt Growth Rate as Sufficient Statistic

The lesson so far is that first, the price level is determined using the asset-market clearing condition. Second, that the growth rate of nominal debt determines the inflation rate, which together with the nominal interest rate set by monetary policy, determines the real interest rate and thus household asset demand. To explain these lessons, I made the pedagogical simplifying assumption that fiscal policy is specified in nominal terms and sets a constant growth rate of nominal debt. This assumption also allows for a clear explanation in Section 3.5.1 of why the DTPL is different from the FTPL. In the remainder of this section, I relax this assumption and allow for richer nominal and real tax rules and show that the lessons of the previous sections carry over. First, the price level is still determined using the asset-market clearing condition. Second, the growth rate of nominal debt is a “sufficient statistic” which is again sufficient to determine the real interest rate and thus household asset demand. The difference now is that the debt growth rate is not set directly but indirectly through the tax rule. I start with fiscal rules specified in nominal terms in Section 3.5.2 before moving to fiscal rules in real terms in Section 3.5.3. Sections 3.5.4 and 3.5.5 use these insights to discuss the approaches in Kaplan et al. (2023) and Brunnermeier et al. (2023), respectively.

3.5.1 | FTPL

The FTPL¹⁸ takes a different route to determine the steady-state price level. The starting point is the government budget constraint in a complete markets model, which in steady state is

$$\frac{B}{P}(1+i) = \sum_{t=0}^{\infty} \beta^t s = \frac{s}{1-\beta} \quad (26)$$

for a real primary surplus s and using the real interest rate as $1 + r = 1/\beta$. In the model in Section 2, the real primary surplus $s = \tau$ for real lump-sum taxes τ .¹⁹ The solution for the price level is then

$$P^{FTPL} = \frac{B(1+i)(1-\beta)}{s} = \frac{B(1+\pi)r}{s}, \quad (27)$$

which simplifies to $Ps = Br$ if $\pi = 0$. As explained in Section 3.4, one way to understand why the price level in complete markets models is indeterminate is to note that the number of endogenous variables exceeds the number of equilibrium conditions by one. The FTPL provides an additional equation, as it assumes that the government budget constraint is satisfied by only one price level. In complete markets, this is equivalent to the assumption that the household transversality condition,

$$\lim_{T \rightarrow \infty} \beta^T u(c_T) \frac{B_{T+1}}{P_T} = 0$$

is satisfied by only one price level, P^{FTPL} . If, for example, $P_0 < P^{FTPL}$, then real debt eventually grows at the interest rate $1/\beta$ and the transversality condition does not hold.

While this is not the approach pursued in this paper, there is a similarity. The FTPL uses an equation which states that nominal debt divided by the price level is equal to “something real,” the discounted present value of primary surpluses. This paper uses a different equation—the asset-market clearing condition—which also states that nominal debt divided by the price level is equal to “something real,” namely, real aggregate asset demand.

There are, however, two reasons why the two theories are different and why the FTPL does not operate here.²⁰ First, the government budget is specified in nominal terms, so that the nominal primary surplus equals $\Sigma = T$ for nominal lump-sum taxes T ($\Sigma = T - G$ if nominal government expenditures $G > 0$). The government budget constraint in real terms is

$$\frac{B}{P} = \frac{\Sigma}{P(1-\beta)}. \quad (28)$$

Since the surplus is in nominal and not in real terms, the price level shows up on both sides of the equation, preventing us from solving the equation for P . The government budget constraint now has to be satisfied in nominal terms,

$$B = \frac{\Sigma}{1-\beta}, \quad (29)$$

and if it is, the corresponding real constraint is then satisfied for all price levels P , rendering P indeterminate.

A second difference is that here, taxes T are adjusted to balance the government budget at all times, taking prices as given. In contrast, the FTPL assumes the government does not take prices as given when choosing B and s , so that nominal bonds and the real surplus are chosen first and then prices adjust. More generally, the FTPL requires fiscal policy to be active (Leeper 1991), that is, the fiscal authority does not adjust taxes to balance the budget at all times. Without this assumption, fiscal policy is passive, taxes are adjusted to balance the budget, and the FTPL is not operating and the equation stating that nominal debt divided by the price level is equal to “something real” is not well-defined. Since this paper assumes a passive fiscal policy, the price level is determinate but not according to the FTPL. A further discussion of the differences between my theory and the FTPL using Ljungqvist and Sargent’s (2012) “Ten Monetary Doctrines” environment is provided in Appendix B.²¹

3.5.2 | Nominal Tax Rules

I assume that nominal taxes are set according to the rule,

$$T_t = \omega_1 i_t B_t + \omega_2 B_t, \quad (30)$$

for parameters ω_1 and ω_2 , and show how this indirectly sets the nominal debt growth rate. Substituting the tax rule into the flow budget constraint,

$$B_{t+1} = (1+i_t)B_t - T_t = (1-\omega_1)i_t B_t + (1-\omega_2)B_t \quad (31)$$

yields the nominal debt growth rate,

$$\frac{B_{t+1}}{B_t} = (1 - \omega_1)i_t + (1 - \omega_2), \quad (32)$$

such that the steady-state inflation rate is

$$(1 + \pi_{ss}) = (1 - \omega_1)i_{ss} + (1 - \omega_2). \quad (33)$$

Setting a debt growth rate of $x\%$ as in the previous sections arises now for $\omega_1 = 1$ and $\omega_2 = -x$, and all previous results obviously hold. Monetary policy has no effect on the inflation rate, since $\omega_1 = 1$ means that fiscal policy adjusts taxes in proportion to interest rate costs. Subject to these choices, fiscal policy can fully control the steady-state inflation rate. If instead $\omega_1 \neq 1$ changes in i_{ss} affect the growth rate of debt and thus the inflation rate. If $\omega_1 < 1$, a higher nominal interest rate i_{ss} means higher debt growth and thus higher inflation, since the higher interest rate costs are only partly covered by higher taxes. On the other hand, $\omega_1 > 1$, a higher nominal interest rate i_{ss} means lower debt growth and thus lower inflation, since taxes increase more than the interest rate costs.

3.5.3 | Real Tax Rules

I now assume that taxes are set in real terms according to the rule

$$\tau_t = \tau^* + \gamma(r_t b_t - \tau^*), \quad (34)$$

where τ_t is Period t taxes, $\tau^* > 0$, real bonds are defined as $b_t = B_t/P_{t-1}$ and either $\gamma \in [0, 1)$ (an active policy in Leeper 1991) or $\gamma > 1$, a passive fiscal policy. Substituting into the government budget constraint

$$B_{t+1} = (1 + i_t)B_t - P_t \tau_t \quad (35)$$

$$= (1 + i_t)(1 - \gamma)B_t + \gamma B_t(1 + \pi_t) - P_t(1 - \gamma)\tau^*, \quad (36)$$

allows deriving the growth rate of debt,

$$\frac{B_{t+1}}{B_t} = (1 + i_t)(1 - \gamma) + \gamma(1 + \pi_t) - \frac{P_t}{B_t}(1 - \gamma)\tau^* \quad (37)$$

and the steady-state inflation rate if $\gamma \neq 1$,²²

$$(1 + \pi_{ss}) = 1 + i_{ss} - \frac{P_t}{B_t}\tau^*. \quad (38)$$

Fiscal and monetary policy now do not uniquely determine the growth rate of nominal debt and thus inflation. But the price level is still determined in the asset market, as the left panel (a) of Figure 3 illustrates. It is convenient to operate in the price-asset space, since the inflation rate depends on the price level. If the primary surplus $\tau^* > 0$ and the asset demand curve is upward sloping, then $S((1 + i_{ss})/(1 + i_{ss} - \tau^*P/B))$ is increasing in P and we obtain a unique asset-market clearing equilibrium with a unique price level P^* and a unique inflation rate $\pi_{ss} = i_{ss} - \tau^*P^*/B$. This case is shown in panel (a). If fiscal policy instead follows a nominal rule, for example, setting a constant nominal debt growth rate, then the price level is unique, independently of

the properties of the asset demand function $S(1 + r)$. In particular, monotonicity is not required for uniqueness.

This uniqueness result does not, however, extend to real tax rules. If the $S(1 + r)$ curve is downward-sloping in $1 + r$ or the surplus is negative, $\tau^* < 0$, then $S((1 + i_{ss})/(1 + i_{ss} - \tau^*P/B))$ is downward-sloping in P , as panel (b) shows (for $\tau^* < 0$), and we obtain two steady-state price levels. Both price levels clear the asset market, but the real tax rule does not uniquely determine the growth rate of nominal debt. The two steady states are associated with different growth rates of nominal debt, π^a and π^b . A simple way to see this is merely to recognize that fiscal policy can select one of the two steady states through directly implementing a constant debt growth rate, in which debt grows at rate π^a or π^b . Following this strategy leads to a unique steady state, since we are back in the nominal tax rule scenario. Interestingly, the results are identical for all values of γ and thus independently of whether fiscal policy is active or passive.

3.5.4 | Approach in Kaplan et al. (2023)

Kaplan et al. (2023) (KNV) aim at integrating the FTPL into a heterogeneous agents incomplete markets model. They use a real tax rule as in (34) and the previous section shows how this determines the growth of nominal debt, the inflation rate, and the price level. However, KNV suggest a different approach to price-level determination. To understand their approach, recall that the FTPL determines the price level through the present-value government budget constraint, which is obtained from iterating the flow budget constraint, noting that $\tau^* > 0$,

$$\frac{B_0}{P_0}(1 + i_{ss}(1 - \gamma)) = \sum_{j=0}^{\infty} \left(\frac{1}{1 + r_{ss}(1 - \gamma)} \right)^j \tau^* + \lim_{t \rightarrow \infty} \frac{b_{t+1}}{(1 + r_{ss}(1 - \gamma))^t} = \begin{cases} < \infty, & \text{for } \gamma < 1 \\ = \infty, & \text{for } \gamma > 1. \end{cases} \quad (39)$$

The right-hand side is exogenous and finite if $\gamma < 1$ and thus determines the initial price level, restating the finding in Leeper (1991) that an active fiscal policy ($\gamma < 1$) delivers determinacy. An active fiscal policy delivers the additional equation, needed to determine the price level. However, the right-hand side is infinite if a passive fiscal policy ($\gamma > 1$) is pursued and one does not obtain the necessary additional equation, and thus the price level is indeterminate.

KNV deviate from this approach and use the flow instead of the present-value government budget constraint. This is clearly not the FTPL since it means that the government neither raises taxes nor issues new debt to balance the flow budget constraint. Instead, the price level has to adjust to ensure a balanced budget in each period.

As explained above, price-level determinacy requires an additional equation. KNV use the flow budget constraint in steady state as if it constituted the necessary additional equation. This approach cannot be correct though since it would also erroneously deliver price-level determinacy for complete markets. The government flow budget constraint without time indices reads

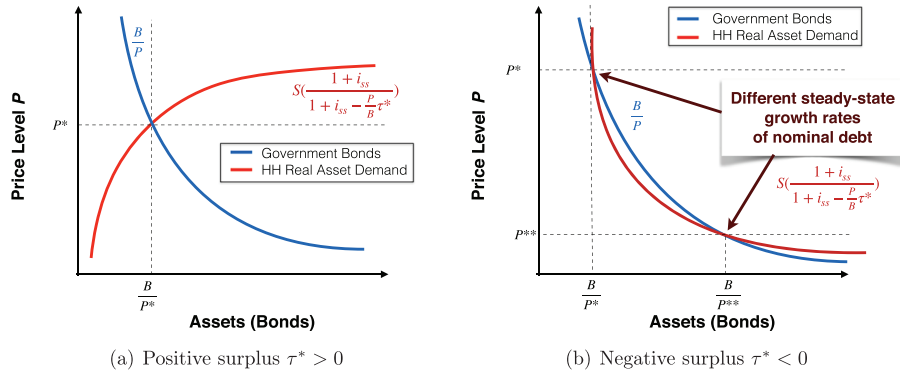


FIGURE 3 | Asset market equilibrium with real tax rule (34), $\tau_t = \tau^* + \gamma(r_t b_t - \tau^*)$.

$$r_s b_{ss} = \frac{1 - \beta}{\beta} \frac{B}{P^*} = \tau^*, \quad (40)$$

which yields the price level P^* for all values of γ . The reason is that the flow budget constraint is used as if it was an additional equation, although we know that an equation is missing in complete markets models if fiscal policy is passive ($\gamma > 1$).

The reason for this incorrect implication is that KNV first move to stationary values and then obtain an equation which seems to determine the price level in steady state, although the price level is not determinate if time indices are not dropped. The correct approach would be (as in this paper or the FTPL) to firstly write the equilibrium equations with a time index including the additional equation which determines the price level. Then secondly, drop the time indices and replace the variables with their stationary values.

Although the KNV approach of using the flow budget constraint can be incorrect in complete markets models, it delivers the correct answer in incomplete markets models. The reason is that the KNV approach applied to incomplete markets models implicitly does what I described in the previous sections. My approach starts with household asset demand, which depends on current assets and future prices, interest rates, taxes, and income. The price level is then determined from asset market clearing, which equates household asset demand and asset supply. The fiscal tax rule affects household asset demand and determines the growth rate of nominal government debt, which in turn determines the supply of nominal government bonds. None of these arguments requires a steady state, as the asset-market clearing condition features time indices. I then move to a steady state, in which the growth rate of nominal debt is constant, equal to the inflation rate and satisfies the steady-state flow government budget constraint.

This is why the KNV approach delivers the correct result, simply because it delivers the correct steady-state growth rate of nominal debt. But this growth rate is merely the input into the asset-market clearing condition which ultimately determines the price level. This shows why the price level is determinate in incomplete markets models, but not if markets are complete, although both models feature a flow government budget constraint and thus KNV would work in both models. This holds because the asset-

market clearing condition is an additional equation and thus determines the price level in incomplete-markets, but not in complete-markets models (see Figure 2). The tax rule and the flow government budget constraint are merely inputs with which to compute household asset demand and do not constitute an additional equation.

3.5.5 | Price-Level Determinacy in Brunnermeier et al. (2023)

Brunnermeier et al. (2023) (BMS) also aim at incorporating the FTPL into incomplete markets models. The economic mechanism operating in their model is again the one described in the previous sections. The argument is straightforward, since BMS use the same specification as described in Sections 3.1–3.3. The growth rate of nominal debt is exogenous at rate μ_t and thus constant in a steady state at exogenous rate μ . Taxes are set to balance the flow government budget constraint,

$$P_t \tau_t := (1+i)B_t - B_{t+1} = (1+i)B_t - (1+\mu_{t+1})B_t = (i - \mu_{t+1})B_t \quad (41)$$

rendering fiscal policy passive. The previous sections showed how the unique finite price level is determined under these assumptions.

Instead of following the logic of the DTPL, BMS suggest that ensuring a unique present value of government surpluses is sufficient for price-level determinacy. They describe the FTPL as equating the real value of debt to the present value of surpluses independently of whether this equation constitutes an additional equation needed for price-level determinacy or not. The steady-state version of their definition using my notation reads

$$\frac{B_0}{P_0}(1+i_{ss}) = \sum_{j=0}^{\infty} \left(\frac{1}{1+r_{ss}} \right)^j \tau_{ss}. \quad (42)$$

However, as explained above, price-level determinacy requires an additional equation. In the FTPL, the present-value government budget constraint is used, but it constitutes an additional equation only if fiscal policy is active. Deriving this additional equation requires iterating the flow budget constraint, which yields (39), showing that policy has to be active, $\gamma < 1$, to obtain an additional equation. If policy is passive, $\gamma > 1$, the

present-value government budget constraint is always satisfied as an equilibrium requires and thus does not constitute an additional equation.

In BMS, different price levels constitute an equilibrium and are associated with different present values of government surpluses. Indeed, Equation (42) is satisfied for any price level P_0 ,

$$\begin{aligned} \frac{B_0}{P_0}(1 + i_{ss}) &= \sum_{j=0}^{\infty} \left(\frac{1}{1 + r_{ss}} \right)^j \tau_{ss} = r_{ss} b_{ss} \frac{1 + r_{ss}}{r_{ss}} \\ &= (1 + r_{ss}) \frac{B_t}{P_t} (1 + \pi_{ss}) = \frac{B_0}{P_0} (1 + i_{ss}), \end{aligned} \quad (43)$$

using the fact that the passive fiscal rule (41) in steady state equals

$$\tau_{ss} = (i_{ss} - \pi_{ss}) \frac{B_t}{P_t} = \frac{i_{ss} - \pi_{ss}}{1 + \pi_{ss}} \frac{B_t}{P_{t-1}} = r_{ss} b_{ss}. \quad (44)$$

Equation (43) is a trivial identity, which does not determine the price level P_0 , which cancels out of this equation. These arguments apply equally to complete and incomplete markets models.

The difference between incomplete and complete markets is that the price level is nevertheless determinate in the incomplete markets model, but according to the theory laid out in the previous sections. Since the price level is determined through the asset-market clearing condition, we obtain unique values for all variables including the present value of government surplus. This present value is unique, since the price level is uniquely determined according to the DTPL. In light of the trivial identity (43), it is incorrect to infer price-level determinacy from the uniqueness of the present value.²³ In addition, as explained above, if this were correct, it should also be correct in a complete markets model, but we know it is wrong there. The reason for this discrepancy between incomplete and complete markets models is that the DTPL works in the former but not in the latter model class.

In incomplete markets models, the FTPL is not equivalent to asset or goods market clearing. The asset market, or equivalently the goods market clearing conditions, always operate. They determine the price level in incomplete markets models since “government debt matters for real quantities” and thus these market clearing conditions depend on the price level. In contrast, the FTPL does not operate if fiscal policy is passive, and therefore cannot be equivalent to asset or goods market clearing.

3.6 | Adding Capital in Production

The determinacy result also holds in a model with investment I_t and capital K_t . To see this, assume a production function $Y_t = F(K_t, h_t) = F(K_t, 1)$ where labor is inelastically supplied, $h_t = 1$, and capital accumulates as

$$K_{t+1} = F(K_t, h_t) + (1 - \delta)K_t - \int c_t d\Omega_t \quad (45)$$

for a depreciation rate δ . Households rent their labor services $h_t e_t = e_t$ to firms for a real wage $w_t = F_h(K_t, 1)$ such that the

budget constraint changes to

$$c_t + a_{t+1} = (1 + r_t)a_t + w_t e_t - \tau_t. \quad (46)$$

Bonds and capital are perfect substitutes, so that asset market clearing requires

$$K_{t+1} + \frac{B_{t+1}}{P_t} = \int a_{t+1} d\Omega_t = S_{t+1}, \quad (47)$$

and both assets have the same return,

$$F_K(K_{t+1}, 1) + (1 - \delta) = 1 + r_{t+1} = (1 + i_{t+1}) \frac{P_t}{P_{t+1}} \quad (48)$$

showing that capital $K_{t+1}(1 + r_{t+1})$ is a function of the real interest rate. The definition of the steady-state asset demand function S is as in Section 2.4, with the only difference being that lump-sum taxes are levied only to cover the interest rate payments on bonds which in equilibrium are equal to $B_t/P_{t-1} = S_t - K_t$, so that now

$$\tau_{ss} = r_{ss}(S_{ss} - K_{ss}). \quad (49)$$

In a steady state, the two conditions (47) and (48) are

$$K^* + \frac{B}{P^*} = S(1 + r_{ss}) = S \left(\frac{1 + i_{ss}}{1 + \pi_{ss}} \right), \quad (50)$$

$$F_K(K^*, 1) + (1 - \delta) = 1 + r_{ss} = \frac{1 + i_{ss}}{1 + \pi_{ss}}, \quad (51)$$

which together determine the steady-state values of the two endogenous variables K^* and $P^* \in \mathbb{R}_{>0}$.²⁴ In particular, the price level is determinate using again the asset-market clearing condition, but now taking into account that there are two assets, bonds and capital. The asset-market clearing condition again provides the additional equation needed for price-level determinacy, while the second equation is used only to determine the capital stock.

3.7 | Money Demand, Endogenous Money, and Open-Market Operations

I now show that the determinacy result derived so far in a cashless economy extends to models where households have a nontrivial demand for money. To generate a demand for money, I assume preferences over consumption c_t and real money balances m_t

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (u(c_t) + \mu(m_t)), \quad (52)$$

where μ is increasing and concave. A household carries nominal money $P_{t-1}m_t$ into Period t from the previous period and acquires money $P_t m_{t+1}$ in that period. Households then maximize utility for a budget constraint

$$c_t + a_{t+1} + m_{t+1} = (1 + r_t)a_t + m_t \frac{P_{t-1}}{P_t} + e_t - \tau_t \quad (53)$$

subject to the credit constraint (2) and $m_{t+1} \geq 0$. Aggregate steady-state assets and real money are

$$S = \int a d\Omega \quad \text{and} \quad L = \int m d\Omega, \quad (54)$$

where Ω is now the distribution on agents' assets a , real balances m , and endowment e across the population. Both S and L are functions of the real interest rate $1 + r$, the inflation rate $1 + \pi$, and taxes τ ,

$$S = S(1 + r, 1 + \pi, \tau) \quad \text{and} \quad L = L(1 + r, 1 + \pi, \tau). \quad (55)$$

Using that steady-state seigniorage revenue is $L\pi/(1 + \pi)$ and thus $\tau = rS - L\pi/(1 + \pi)$ determines asset demand and money as fixed points to²⁵

$$S = S(1 + r, 1 + \pi, rS - L\frac{\pi}{1 + \pi}) \quad \text{and} \quad L = L(1 + r, 1 + \pi, rS - L\frac{\pi}{1 + \pi}) \quad (56)$$

and yields steady-state asset demand and money as a function of the real interest rate and the inflation rate

$$S = S(1 + r, 1 + \pi) \quad \text{and} \quad L = L(1 + r, 1 + \pi). \quad (57)$$

The central bank adjusts money supply M to satisfy whatever money demand households have given the nominal interest rate $i_{ss} > 0$ set by the central bank,²⁶ rendering M an endogenous variable.²⁷ The steady-state price level P^* and money M^* are determined as solutions to

$$\frac{M^*}{P^*} = L\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right), \quad (58)$$

$$\frac{B}{P^*} = S\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right). \quad (59)$$

Now there are two equations in two unknowns M^* and P^* , where B and π_{ss} are set by fiscal policy and i is set by monetary policy. The central bank has to provide nominal money

$$M^* = P^* \int m d\Omega = P^* L\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right) \quad (60)$$

to implement the nominal interest rate $1 + i_{ss}$. Here, the assumption is that households exchange consumption goods for money. If one assumes instead that households obtain money through open-market operations, then P^* and M^* solve

$$\frac{M^*}{P^*} = L\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right), \quad (61)$$

$$\frac{B - M^*}{P^*} = S\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right), \quad (62)$$

again with two equations in two unknowns, or equivalently P^* solves

$$\frac{B}{P^*} = S\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right) + L\left(\frac{1 + i_{ss}}{1 + \pi_{ss}}, 1 + \pi_{ss}\right). \quad (63)$$

Clearly, Equation (61) alone does not determine the price level, since the central bank sets i and not M , which adjusts endogenously to satisfy the quantity equation. Instead it is the asset-market clearing condition that determines the price level, taking fiscal and monetary policy variables as given.

Both the real value of money M and of bonds B are constant in a steady state,

$$1 + \pi_{ss} = \frac{B' - B}{B} = \frac{M' - M}{M}. \quad (64)$$

With open-market operations, households hold $B - M$ nominal bonds and money M in a steady state, which both grow at the same rate as the supply of nominal bonds B , which again equals the inflation rate:

$$1 + \pi_{ss} = \frac{B' - B}{B} = \frac{(B - M)' - (B - M)}{B - M} = \frac{M' - M}{M}. \quad (65)$$

This is a steady-state condition which will hold in any model with government bonds and money.²⁸ In simple textbook models, the central bank sets the money supply and according to the standard interpretation, determines the steady-state inflation rate equal to the growth rate of money. If the central bank sets M'/M to 2% then the inflation rate equals 2% in a steady state. If, in another steady state, M'/M equals 4%, then the inflation rate equals 4%. Fiscal policy then has no choice but to adjust the growth rate of bond supply to be equal to the inflation rate determined by monetary policy. Causality runs from M'/M to π_{ss} to B'/B . What is different here is that money is not a policy instrument, but adjusts to money demand and price movements, so that the causality runs from π_{ss} to M'/M . Fiscal policy sets B'/B which is then equal to the steady-state inflation rate, so that the causality runs from fiscal policy to inflation. Again, this is a comparison of steady states, which first requires fiscal policy to be consistent with steady-state conditions, since otherwise the economy is not in steady state. And secondly, this adds a causal interpretation, as fiscal policy follows a simple constant debt-growth-rate policy. If this simple rule sets B'/B equal to 2%, then the inflation rate equals 2%, and setting B'/B equal to 4% implies an inflation rate of 4%. In contrast to the FTPL, no game between the fiscal and the monetary authority needs to be specified. The result here is just a combination of a steady-state condition and the fact that fiscal policy implements a constant debt growth-rate policy, and monetary policy a constant nominal interest rate. These two policies are consistent, as all equilibrium conditions are satisfied, so that no game needs to be specified to resolve any inconsistencies.

In this sense, fiscal policy can determine the long-run inflation rate (and the price level) through commitment to such a simple rule.²⁹ One interpretation of determinacy is that causality runs from policy to prices and inflation and that reverse causality is ruled out. An example of indeterminacy and thus reverse causality would be a policy which keeps the nominal interest rate and real bonds constant. The steady-state equation $1 + \pi_{ss} = (B' - B)/B$ would still hold, but would be driven by inflation sunspots fully accommodated by fiscal policy through adjustments in nominal bonds.

3.8 | Price-Level Determinacy With Nominal Government Expenditures

The basic model in the previous sections links the inflation rate and the price level to the supply of nominal government bonds. I now argue that the DTPL offers a broader perspective on price determination than this simple link operating in the basic model. In the DTPL, the price level and the inflation rate are linked to nominal aggregate demand which in turn is derived from nominal government spending, bonds, taxes, and transfers.

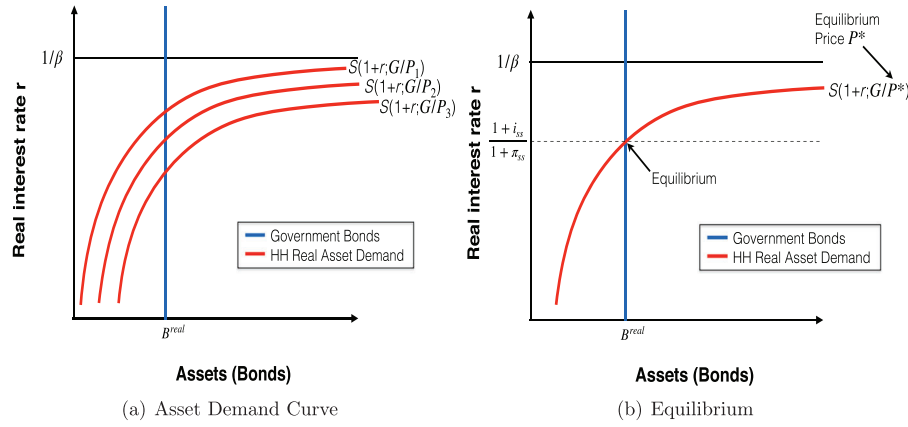


FIGURE 4 | Asset market equilibrium with price-indexed government debt B^{real} .

I therefore first generalize the discussion of price-level determinacy to a scenario in which government bonds are real, and show that the price level is nevertheless determinate if spending and taxes are (partially) nominal. For illustrative purposes, I consider the extreme case of the real value of government bonds fixed at B^{real} , and government spending and taxes entirely nominal.³⁰ One example is an economy without government bonds, $B^{real} = 0$, and where nominal taxes are equal to nominal government spending in each period.

Period t nominal government expenditure is G_t so that the nominal government budget constraint equals

$$T_t := (1 + i_t)P_{t-1}B^{real} + G_t - P_t\omega - P_tB^{real} \quad (66)$$

for a linear endowment tax ω . This linear tax is not essential and its only purpose is rendering transfers $T < 0$ possible. Households' budget constraints change to

$$c_t + a_{t+1} = (1 + r_t)a_t + (1 - \omega)e_t - \tau_t. \quad (67)$$

In a steady state, real tax revenue and real government expenditure are constant, such that the steady-state inflation rate is again determined by fiscal policy

$$1 + \pi_{ss} = \frac{T' - T}{T} = \frac{G' - G}{G}. \quad (68)$$

As shown in Section 2.4, steady-state asset demand $S(\Omega_{ss}, 1 + r_{ss}, 1 + r_{ss}, \dots; \tau_{ss}, \tau_{ss}, \dots)$ depends on the real interest rate, $1 + r_{ss}$, and the real value of taxes, $\tau_{ss} = T/P$, and is now denoted $S(1 + r_{ss}, \tau_{ss})$. The difference is that now taxes are raised to pay for government expenditure and interest rate payments, implying that the real value of taxes and thus asset demand depends on the price level and the asset-market clearing condition is

$$S = S(1 + r_{ss}, G/P) := S(1 + r_{ss}, \frac{G}{P} - \omega + r_{ss}B^{real}) = B^{real}, \quad (69)$$

where taxes $T/P := G/P - \omega + rB^{real}$.

Result 1 in Appendix A.1 establishes that the steady-state asset demand S is strictly decreasing in $\tau_{ss} = T/P$ and thus the function S is strictly decreasing in G/P . The left panel of Figure 4 illustrates this result as different price levels shift the asset demand curve, while the real value of bonds is unchanged at B^{real} .

The reason why household real asset demand depends on the level of real taxes, T/P , for a fixed real interest rate, is explained by heterogeneity and incomplete markets, which in turn imply failure of the PIH and that agents, as a result of this failure, engage in precautionary savings. A lower steady-state level of real taxes increases both demand and (precautionary) savings. This reasoning extends to changes in the price level, which translate one-for-one into changes in real taxes, since the nominal level of taxes is fixed. The intuition is straightforward. A higher steady-state price level lowers real government consumption, since fiscal policy is nominal, and at the same time, lowers the tax burden for the private sector by the same amount. Households, however, do not spend all of the tax rebate on consumption, but instead, use some of the tax rebate to increase their precautionary savings. This less than one-for-one substitution of private sector demand for government consumption implies a drop in aggregate demand (households plus government demand) and an increase in household savings. This would require an adjustment of the real interest rate so as to stimulate demand and lower savings, such that both the goods and the asset markets clear. The steady-state real interest rate cannot adjust to equate supply and demand, because it is pinned down by the nominal interest rate set by monetary policy, and by the inflation rate, which is equal to the growth rate of nominal government spending. Therefore, the equilibrium price level P^* must adjust such that demand equals supply when the real interest rate equals $1 + r_{ss} = (1 + i_{ss})/(1 + \pi_{ss})$.

Existence of a finite steady-state price level requires that the private sector is willing to absorb B^{real} given the real interest rate determined by fiscal and monetary policy

$$S(\frac{1 + i_{ss}}{1 + \pi_{ss}}, \tau = r_{ss}B^{real} - \omega) > B^{real} \geq S(\frac{1 + i_{ss}}{1 + \pi_{ss}}, \bar{\tau}), \quad (70)$$

where $\bar{\tau}$ is the largest feasible lump-sum tax.³¹ Continuity in τ follows from (Açikgöz 2018) or Achdou et al. (2022), so that the Intermediate Value Theorem and monotonicity imply existence of a unique finite steady-state price level $P^* \in \mathbb{R}_{>0}$ solving

$$S(\frac{1 + i_{ss}}{1 + \pi_{ss}}, G/P^*) = B^{real}, \quad (71)$$

as the right panel of Figure 4 illustrates. Note that uniqueness again refers to the price level for a given real interest rate $1 + r_{ss}$.

and does not rule out multiple real steady states with different real interest rates. If condition (70) does not hold, a steady state might not exist for the chosen real interest rate.

In the general case in which government bonds are nominal and $G_t > 0$, the nominal government budget constraint equals

$$T_t := (1 + i_t)B_t + G_t - P_t\omega - B_{t+1}. \quad (72)$$

Steady-state asset demand $S(1 + r, G/P)$ is then a fixed point of

$$S = S(1 + r, \frac{G}{P} - \omega + rS) \quad (73)$$

resulting in the steady-state asset-market clearing condition

$$S(1 + r, \frac{G}{P}) = \frac{B}{P}. \quad (74)$$

Previous arguments carry over and imply that asset demand S is continuous and strictly decreasing in G/P and thus S is continuous and strictly increasing in P . Imposing

$$S(\frac{1 + i_{ss}}{1 + \pi_{ss}}, \tau = -\omega) > 0 \quad \text{and} \quad B/\bar{P} \geq S(\frac{1 + i_{ss}}{1 + \pi_{ss}}, \bar{\tau}) \quad (75)$$

now ensures existence of a unique finite steady-state price at which the private sector is willing to absorb B/P given the real interest rate determined by fiscal and monetary policy, and $\bar{\tau}$ is the largest feasible lump-sum tax and equals $\bar{\tau} = G/\bar{P} + rB/\bar{P} - \omega$ for a price level $\bar{P}$.

4 | Concluding Remarks

This paper shows that the steady-state price level is determinate in Bewley–Imrohorglu–Huggett–Aiyagari heterogeneous agents incomplete markets models. A key finding is that the price level is determined jointly by monetary and fiscal policy, with long-run inflation determined by the growth rate of nominal fiscal variables—nominal government expenditure, nominal taxes and transfers, and government debt—even if monetary policy is operating an interest rate rule with a different inflation target. The nominal anchor—nominal fiscal variables—is controlled by fiscal policy, which therefore has the power to set the long-run inflation rate.

Price-level determinacy is derived within the standard incomplete markets framework, with a partially nominal fiscal policy and without any additional new assumptions. The determinacy result is then merely a consequence of well-known properties of aggregate consumption, savings, and real interest rates in incomplete markets models. The literature has, however, not recognized how combining these properties yields determinacy.

My theory establishes a new link between fiscal policy and inflation rates. The empirical evidence in Sargent et al. (2009) and Esquivel et al. (2019) suggests that this fiscal link operates in developing countries and not only during hyperinflations. High inflation rates are caused by expansionary nominal fiscal policies.

For developed countries, the evidence is less obvious. Coibion et al. (2021) show, for the United States, consistent with my theory,

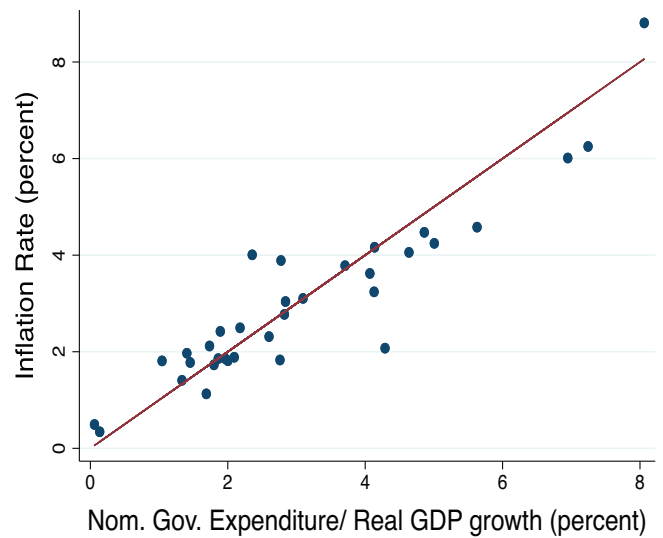


FIGURE 5 | Inflation and nominal government expenditure.

Note: Government expenditure = Final consumption expenditure (GP3P) + Gross capital formation and Acquisitions less disposals of nonassets (GP5_K2P) – Consumption of fixed capital (GK1R), corresponding to the definition in Hagedorn et al. (2024b) using line numbers (21, 39, 41, 42) in NIPA Table 3.1. Inflation is calculated using the OECD CPI Index for all items. All averages for each country are calculated for the period where both data series are available.

that short- and long-run inflation expectations respond to news about future fiscal policy, but not to current debt or deficit levels. One prediction of my theory is that the growth rate of the ratio of nominal government expenditure to real GDP moves one-for-one with the inflation rate over longer time horizons (Equation 68).³² Figure 5 provides supporting evidence for this prediction, similar to the exercise in McCandless and Weber (1995) and Lucas (1996).

It plots the average annual inflation rates against the average annual growth rate of nominal government expenditure to real GDP for 34 OECD countries.³³ The figure shows that the points are close to the 45° line with a simple correlation of 0.93, as predicted by the theory. The correlation coefficient 0.93 is of the same magnitude as the correlation between money growth and inflation reported in McCandless and Weber (1995) and Lucas (1996): 0.95 for $M2$, 0.96 for $M1$, and 0.92 for $M0$. Clearly, these correlations are supporting evidence for the respective theories, but do not establish causality, that is, whether changes in government expenditure or money supply cause inflation. This result is not conclusive on whether a causal link exists in which fiscal policy increases nominal spending, and this decision causes inflation, or whether nominal government expenditure responds to inflation to keep the real value roughly constant.

Hagedorn (2021) is a first step in establishing such a causal link. I show that both nominal fiscal transfers and marginal costs are significant determinants of inflation. I build on Hazell et al. (2022) and use a panel data set for US states to overcome identification issues commonly encountered in a time-series analysis. My empirical findings support a nominal demand-augmented Phillips curve (NDPC), which adds nominal demand

as a second determinant to a standard New Keynesian Phillips curve (NKPC). In particular, the NDPC features a causal link from fiscal policy to nominal spending and then to inflation.

Certainly, this evidence is only a first step in understanding inflation in the data. Any truly satisfactory explanation should however be based on a unifying theory which can at the same time explain inflation in developing and developed countries, in low-, medium-, and high-inflation countries and during different historical episodes. The theory proposed in this paper, linking inflation to nominal fiscal policy, offers such a unifying explanation.

The novel theory of price and inflation determination also calls for a rethinking of various issues in monetary economics. After the financial crisis in 2008, the inflation rate moved very little, while at the same time, bank reserves exploded as a result of quantitative easing (Cochrane 2018). In contrast, textbook monetarist analysis in which reserves are a nominal anchor, would predict a large increase in inflation in response to such a large increase in reserves, questioning the effectiveness of the monetarist quantity theory channel in generating inflation. This paper offers a new potential explanation for the observed low inflation rate, namely, fiscal policy which did not induce a sufficiently high growth rate of nominal aggregate demand to generate a higher inflation rate.

The theory set out in this paper also offers a different perspective on US inflation history. After experiencing high rates in the 1970s, the 1980s witnessed success in keeping inflation low. The standard interpretation is that a change in monetary policy led to high unemployment which in turn, through a (steep) Phillips curve, induced a rapid fall in inflation. Hazell et al. (2022) challenge this conventional wisdom and instead conclude that the decline in inflation was largely the result of a decline in long-run inflation expectations. This conclusion then raises the question about the determinants of the long-run inflation rate. The theory proposed in this paper suggests an answer.³⁴ It suggests that it was not in fact the change in monetary policy that kept inflation in check, but rather a shift to a less expansionary fiscal policy during the Reagan administration, perhaps imposed by the prolonged high nominal interest rates set by central banks under chairman Paul Volcker, and the resulting high deficits. An interpretation consistent with the view expressed in Sargent (1982), who documents that the abrupt ending of hyperinflation in Austria, Hungary, Poland, and Germany was the result of “a dramatic change in the fiscal policy regime.”

Building on the findings in this paper, Hagedorn (2016) and Hagedorn et al. (2017;2019;2024b) use incomplete markets models to investigate several hotly debated topics in monetary economics. Not surprisingly, the novel way of thinking proposed in this paper yields different answers from those emerging from the conventional view.

In this research, we consider forward guidance, which refers to the notion that central bank commitment to keeping the nominal interest rate low during a period after the liquidity trap, leads to large output gains even during the liquidity trap. We show that the large output gains from forward guidance do not occur in incomplete markets models. Independently of whether the nominal

interest rate is raised earlier or later than in the benchmark, the output path is almost unchanged. Furthermore, New Keynesian models also imply that technological regress is expansionary in a liquidity trap. Again by contrast, we show that this is not the case in incomplete markets models. Technological progress is expansionary and technological regress contractionary.

Hagedorn (2023) proposes the first methodology for assessing local determinacy in incomplete markets models, complementing the steady-state determinacy analysis in this paper. A main finding is that a constant nominal interest rate already ensures determinacy in contrast to complete markets models which require the interest rates to adhere to the Taylor principle. This finding has important implications for the conduct of monetary policy: It enlarges the choice set for monetary policy as an increase in interest rates is not required to ensure determinacy (anchor inflation).

Applying this determinacy result to a fiscal stimulus, Hagedorn et al. (2017) shows that the government spending multiplier is less than one if the Taylor principle is followed and can be larger than one only if the nominal interest rate is (largely) unchanged in response to the stimulus. The incomplete markets determinacy properties thus overcome the complete markets dilemma of choosing between anchoring inflation and an effective fiscal policy without crowding out consumption. Furthermore, in New Keynesian models, fiscal multipliers in a liquidity trap increase as price stickiness is reduced. By contrast, output declines and so does the fiscal multiplier, if price stickiness is reduced in incomplete markets models. These differences between complete and incomplete markets models are explained through their different determinacy properties (Hagedorn et al. 2017).

Recent work by Hagedorn et al. (2024a;2024b) shows that the redistributive effects of an increase in nominal interest rates are unfavorable to, for example, black households, implying that following the Taylor principle is detrimental to inclusiveness. The incomplete markets determinacy properties again resolve this dilemma: A constant nominal interest rate already ensures determinacy and at the same time contributes to inclusiveness.

Having established a framework with a determinate price level allows for a rigorous study that sheds new light on these and many more important policy and empirical questions.

Acknowledgments

I am very grateful to Iouri Manovskii and Kurt Mitman for their invaluable comments, suggestions and insights. I am also indebted to Tom Sargent for his suggestion to use the framework for “Fiscal-Monetary Theories of Inflation” in chapter 26 of Ljungqvist and Sargent (2012), to clarify the differences between the Fiscal Theory of the Price Level (FTPL) and my theory. I also thank many seminar participants for their insights. I have benefited particularly from discussions with and the comments of David Andolfatto, Manuel Amador, Adrien Auclert, Anmol Bhandari, Jaroslav Borovička, Francisco Buera, John Cochrane, Wei Cui, Wouter Den Haan, Nezih Guner, Jonathan Heathcote, Kyle Herkenhoff, Martin Holm, Luigi Iovino, Greg Kaplan, Moritz Lenel, Ben Moll, Herakles Polemarchakis, Xavier Ragot, Matthew Rognlie, Chris Sims, Kjetil Storesletten, Harald Uhlig, Venky Venkateswaran and Iván Werning. Support from the FRIPRO Grant No. 250617 is gratefully acknowledged.

Endnotes

- ¹ Central bank independence ensures, however, that the treasury cannot impose fiscal policies on the central bank, for example, monetizing its debt to finance the government.
- ² In a simple cash-in-advance (CIA) model, aggregate nominal demand equals exogenous money supply and thus also determines prices and inflation in a CIA model as envisaged by Friedman. In both the incomplete-markets and in the CIA models, inflation is a nominal phenomenon as it is determined by aggregate nominal demand. The difference between my theory and Friedman's view is that this demand has different determinants: money supply according to Friedman, and more generally all factors that drive aggregate demand in my model.
- ³ If bonds are in zero net supply, the growth rate of nominal transfers or equivalently the growth rate of nominal government expenditure, constitute a sufficient statistic for fiscal policy.
- ⁴ Two-Agent New Keynesian (TANK) models are a popular model class, due to their theoretical elegance and tractability (Bilbiie 2008;2020;2025).
- ⁵ See Kaplan and Violante (2018) for a recent review of this emerging HANK literature. Additional references include Gornemann et al. (2012), Kaplan et al. (2018), Auclert (2019), and Lütticke (2021) who study monetary policy in a model with incomplete markets and pricing frictions, but with a different focus, emphasizing and quantifying several redistributive channels of the transmission mechanism of monetary policy, which are absent in standard complete markets models. Earlier contributions are Oh and Reis (2012) and Guerrieri and Lorenzoni (2017), who were among the first to add nominal rigidities to a Bewley–Imrohoroglu–Huggett–Aiyagari model. More recent contributions include McKay and Reis (2016) (Impact of automatic stabilizers), McKay et al. (2016) (Forward guidance), Bayer et al. (2019) (Impact of time-varying income risk), Ravn and Sterk (2017) (Increase in uncertainty causes a recession), Den Haan et al. (2017) (Increase in precautionary savings magnifies deflationary recessions), Auclert and Rognlie (2017) (Inequality and aggregate demand), and Auclert et al. (2024) (Intertemporal Keynesian Cross). While all these papers address issues that are complimentary to my paper, the price level is shown to be determined endogenously in equilibrium only in my paper. In terms of assumptions, the main reason for this difference is that government bonds are nominal here, whereas they are assumed to be real in the cited work. Only Hagedorn et al. (2017) (Fiscal multiplier), Hagedorn et al. (2019) (Forward guidance), and Hagedorn et al. (2024b) (Model estimation) assume nominal bonds.
- ⁶ An alternative assumption to an exogenous credit constraint $\bar{a}$ is to impose natural debt limits (NDL). The Appendix considers this alternative assumption.
- ⁷ It is important to bear in mind that here, as in all the recent literature on monetary economics, the central bank sets the nominal interest rate on short-term bonds and not the money supply, which gives rise to price-level indeterminacy in the first place (Sargent and Wallace 1975).
- ⁸ The government budget constraint is specified here in nominal terms. The only purpose of this specification is that it enables to easily distinguishing this paper's theory from the FTPL as I discuss below (Section 3.5.1). My theory allows for adding real and nominal government expenditures and more complicated real or nominal tax functions (see, e.g., Section 3.8). The theory only requires that fiscal policy be partially nominal, that is, not fully indexed to the price level. I assume realistically that $B_t > 0$, noting that $B_t < 0$ could also be accommodated and $B_t = 0$ is feasible in Section 3.8 with strictly positive nominal government expenditure.
- ⁹ More precisely, the fixed-point problem is to find the fixed point Ω_{ss} of the law of motion $\mathcal{T} : \mathcal{M} \rightarrow \mathcal{M}$ which then satisfies $S_{ss} = \int ad\Omega_{ss} = S(\Omega_{ss}; 1 + r_{ss}, 1 + r_{ss}, \dots; r_{ss} \int ad\Omega_{ss}, r_{ss} \int ad\Omega_{ss}, \dots)$. For existence and uniqueness under standard assumptions, see, for example, Acemoglu and Jensen (2015) and Açıkgöz (2018). Hopenhayn and Prescott (1992) show that the stationary distribution is unique if the mixing condition holds, this means if there is sufficient upward and downward social mobility.
- ¹⁰ Note that this steady-state asset demand function is different from the asset demand function which just aggregates individual household decisions taking prices as given. This is on purpose. Substituting taxes τ_{ss} , using the government budget constraint, allows me to characterize the equilibrium through one equation (13) as a function of one unknown, the real interest rate.
- ¹¹ Price rigidities are relevant for the local determinacy analysis in Hagedorn (2023).
- ¹² For a textbook treatment of incomplete markets models and their steady states, see Ljungqvist and Sargent (2012).
- ¹³ With real economic growth of rate γ , $(1 + \pi_{ss})(1 + \gamma) = B'/B = T'/T$.
- ¹⁴ For example if $\bar{i} = 0.02$, $\phi = 1.5$, debt grows at $\frac{B'-B}{B} = 0.02$ and the inflation target $\pi^* = 0$, steady-state inflation is then 2% and the nominal interest rate equals $i^{ss} = 0.02 + 1.5 * 0.02 = 0.05$. In the (less realistic) case that the inflation target of monetary policy $\pi^* = 0.04$ exceeds the 2% that follows from fiscal policy, the steady-state nominal interest rate equals $i^{ss} = \max(0.02 + 1.5(0.02 - 0.04), 0) = 0$ and inflation is still 2%.
- ¹⁵ The mixing condition, which requires enough upward and downward social mobility, ensures uniqueness of the stationary distribution. Without a unique stationary distribution conditional on the real interest rate, a unique price level would be associated with each stationary distribution. Unsurprisingly, this paper's theory overcomes nominal but not real indeterminacies.
- ¹⁶ If $\bar{a} = 0$ (no borrowing), a sufficient condition for an upward-sloping asset demand curve and thus for a unique equilibrium in real incomplete markets models is that the intertemporal elasticity of substitution is weakly greater than one (Achdou et al. 2022). Without additional assumptions, the equilibrium adjustments of the real interest rate induce income and substitution effects which might lead to a backward-bending asset demand curve. This type of indeterminacy is not an issue here since the real interest rate is fixed and the price level and not the real interest rate adjusts to clear the asset market.
- ¹⁷ The same arguments make clear that targeting $B_{t+1}/(P_t Y_t)$ instead is obviously inconsequential for price-level determinacy.
- ¹⁸ Developed by Leeper (1991), Sims (1994;1997), Woodford (1995;1997;1998a;1998b), Dupor (2000), and Cochrane (1999;2001;2005) building on Sargent and Wallace (1981).
- ¹⁹ In models (as in Section 3.8) where real government spending $g > 0$, the real primary surplus $s = \tau - g$.
- ²⁰ Note also that B_t is an endogenous variable in the FTPL and cannot be set independently from the nominal interest rate i_t (Woodford 1995; Ljungqvist and Sargent 2012), whereas the DTPL allows for independent choices of B_t and i_t .
- ²¹ Kocherlakota and Phelan (1999) (KP) impose a limiting condition, $\lim_{t \rightarrow \infty} (M_t + (1 + i_{t+1})B_t) \prod_{s=1}^{t+1} 1/(1 + i_s) = 0$, on the government budget (Equation 25 in KP) to rule out the FTPL (M is money). In their complete markets model, this condition is equivalent to the household transversality condition which has to be satisfied in any equilibrium. But the KP condition is not equivalent to the transversality condition when markets are incomplete, which, for example, and in contrast to KP, is consistent with a real interest rate below the growth rate of the economy. For example, the KP limiting condition would not be satisfied in a steady state where $i \equiv 0$ (ZLB), constant $M, B > 0$, and nominal taxes equal to nominal government spending, $T = G$, although the government budget constraint and the TVC (5) are clearly satisfied. Therefore, (25) in KP does not apply in incomplete markets models and does not indicate whether or not the FTPL is operating.
- ²² Note that the inflation rate equals the growth rate of nominal debt, keeping the real value of debt constant for all initial price levels, including $P < P^{FTPL}$. Recall that $P < P^{FTPL}$ is ruled out according to the

FTPL in complete markets models since real debt would be unbounded and violate the transversality condition. The complete markets FTPL logic thus does not carry over to incomplete markets models since the inflation rate adjusts to ensure that real debt is bounded.

²³ Of course, one can “read off” the price level from any model equation, once the price level is determined by the DTPL. This equation could be the government present-value budget constraint or the labor market condition that the real wage equals productivity. But clearly, this does not mean that the FTPL operates or that the price level is determined through the labor market clearing condition.

²⁴ The real interest rate is now restricted to satisfy $\beta(1 + r_{ss}) < 1$ and $S(1 + r_{ss}) > K^*(1 + r_{ss})$.

²⁵ Existence again follows from Acemoglu and Jensen (2015) using their extension in Section VII to multidimensional aggregates. Steady-state policy choices of r_{ss} and π_{ss} are restricted such that $0 < S < \infty$ and $0 \leq L < \infty$ to ensure a steady state with a nonzero finite price level.

²⁶ While the model without money allows for $i_{ss} < 0$, adding money precludes negative nominal interest rates but allows for $i_{ss} = 0$ if there is a finite satiation level of money.

²⁷ The same argument holds when the central bank pays an interest rate i^m on money holdings. Again, the central bank has to satisfy household money demand. Paying interest rates on reserves also does not overcome the indeterminacy issue and only changes the opportunity costs of holding money. I thus omit this complication. Diba and Loisel (2021) allow for $i^m > 0$ and in contrast to this paper, assume that the central bank sets money supply M and not the nominal interest rate on bonds i . It is this latter (and not the first) assumption on setting M that delivers determinacy in their model.

²⁸ If the central bank sets $i_{ss} = 0$ and the utility function is such that money demand is finite at $i_{ss} = 0$, households are indifferent between holding bonds and money, leaving the ratio of money to bond holdings indeterminate. This is however irrelevant to the arguments, since $S(\cdot) + L(\cdot)$ is well-defined, the total supply of nominal assets still grows at rate $\frac{B' - B}{B}$ equal to the inflation rate, and the price level is determinate using (63).

²⁹ Hagedorn (2023) considers endogenous fiscal rules which respond to movements in prices and output and characterizes which rules imply determinacy.

³⁰ This assumption again makes clear that the theory in my paper is different from the FTPL, which is discussed in Section 3.5.1 and where the price level is determined such that the real value of bonds clears the government present-value budget constraint. Obviously, the FTPL has no bite, if the real value of bonds is fixed and nominal taxes are set to balance the budget each period.

³¹ Income heterogeneity, credit constraints, and nonnegative consumption imply an upper bound on feasible lump-sum taxes.

³² The corresponding prediction for government debt is more involved, since households savings in the data are divided between various other assets than just government debt, for example, capital. In addition, a substantial fraction of US debt is held abroad.

³³ The 34 countries are: Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea, Latvia, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, Slovak Republic, Slovenia, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States. Data from <http://stats.oecd.org/>.

³⁴ A very flat Phillips curve also raises the question of how inflation can (temporarily) move, when at the same time, inflation expectations do not. A question that might be relevant in the aftermath of the Corona pandemic. This paper offers a possible explanation: A temporary variation in nominal fiscal policy which does not operate through the Phillips curve but nevertheless affects inflation.

³⁵ The Period t intertemporal budget constraint for $e_t = e$ implies

$$(1 + r_{ss})(a_{1,t}(e) - a_{2,t}(e)) = E_t \sum_{s=t}^{\infty} \left(\frac{1}{1 + r_{ss}} \right)^{s-t} \left(\frac{T(e_s)}{P_1} - \frac{T(e_s)}{P_2} \right)$$

and the Period $t + 1$ intertemporal budget constraint for $e_{t+1} = e$ implies

$$(1 + r_{ss})(a_{1,t+1}(e) - a_{2,t+1}(e)) = E_{t+1} \sum_{s=t+1}^{\infty} \left(\frac{1}{1 + r_{ss}} \right)^{s-t-1} \left(\frac{T(e_s)}{P_1} - \frac{T(e_s)}{P_2} \right)$$

so that the differences in beginning of Period t assets, $a_{1,t}(e) - a_{2,t}(e)$, and of Period $t + 1$ assets, $a_{1,t+1}(e) - a_{2,t+1}(e)$, are identical,

$$a_{1,t}(e) - a_{2,t}(e) = a_{1,t+1}(e) - a_{2,t+1}(e),$$

so that these terms cancel in (84).

³⁶ Bénassy (2005;2008) implicitly assumes a particularly strong dynamic inefficiency condition—the population growth rate exceeds the real interest rate (which exceeds $1/\beta$)—since consumption by the initial generation would otherwise eventually exceed GDP.

³⁷ Note that the Period $t - 1$ real value of nominal assets b_t acquired in Period $t - 1$ is b_t/P_{t-1} and that the real Period t payoff is $b_t/P_{t-1}(1 + i) \frac{P_{t-1}}{P_t} = b_t \frac{1+i}{P_t}$, explaining why Period t savings depend on $\frac{1+i}{P_t}$. Household Period t real asset income thus equals $b_t \frac{1+i}{P_t}$ and the household problem is fully real.

³⁸ Computing an equilibrium requires an iteration of two steps. Given a sequence of distributions, we compute the sequence of prices backward. Given a sequence of prices and thus real interest rates, we compute the household problems which yield a new sequence of distributions.

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Appendix A.1: Appendix to Section 3

In this section, I first prove the result on the properties of the asset demand function in the model with positive nominal government expenditure in Section 3.8. I then explain price-level determinacy in incomplete markets models with a natural borrowing constraint (Section A.2). In the remainder of this section, I discuss additional aspects of steady-state price-level determinacy, supplementing the arguments in Section 3, with:

- Explaining price-level indeterminacy in perpetual youth models (Section A.3).
- Explaining price-level indeterminacy in representative agent models with aggregate risk (Section A.4).

A.1.1 | Price-Level Determinacy With Nominal Government Expenditures

This section establishes the result that steady-state asset demand is decreasing in taxes τ .

Result A1. The steady-state asset demand function $S(1 + r, G/P)$ defined in (73) of the model with nominal government expenditure G_t , nominal government bonds, and household budget constraints,

$$c_t + a_{t+1} = (1 + r_t)a_t + (1 - \omega)e_t - \tau_t, \quad (\text{A.1})$$

and government budget constraints,

$$T_t := (1 + i_t)B_t + G_t - P_t\omega - B_{t+1} \quad (\text{A.2})$$

with $\tau_t = T_t/P_t$ is weakly decreasing in G/P . It is strictly decreasing if the coefficient of absolute risk aversion, $\Xi(c) = -u''(c)/u'(c)$, is strictly decreasing or increasing.

Equivalently and using the same proof, $S(1 + r, G/P)$ defined in (69) for the model with real bonds B^{real} is decreasing in G/P .

The proof is largely an application of the results in Acemoglu and Jensen (2015). The steady-state asset demand function $S(1 + r, G/P)$ is the outcome of the fixed-point problem of finding the fixed-point distribution Ω_{ss} of the law of motion $\mathcal{T} : \mathcal{M} \rightarrow \mathcal{M}$, satisfying

$$\begin{aligned} S_{ss} &= \int ad\Omega_{ss} = S(\Omega_{ss}; 1 + r_{ss}, G/P - \omega + r_{ss} \int ad\Omega_{ss}) \\ &= S(1 + r_{ss}, G/P). \end{aligned}$$

Acemoglu and Jensen (2015) establish existence and uniqueness under standard assumptions. Clearly, the steady-state asset demand depends on r_{ss} and G/P .

For comparative statics with respect to lump-sum taxes $\tau = G/P - \omega + r_{ss}S_{ss}$, Acemoglu and Jensen (2015) define a positive shock as a change in an exogenous parameter which leads to an increase in a household's decision variable. I therefore show that an increase in steady-state transfers is a positive shock, since this implies that this increase in steady-state transfers leads to an increase in individual savings in an incomplete markets model.

Lemma 1 in Acemoglu and Jensen (2015) provides sufficient conditions for an exogenous parameter change to be a positive shock, requiring checking the assumptions of Lemma 1 in Acemoglu and Jensen (2015), which builds on Topkis' monotonicity theorem. Acemoglu and Jensen (2015) show that an incomplete markets model satisfies their assumptions 1 and 3.

What remains to be shown (to apply Lemma 1) is that a household's budget constraint has strict complementarities, that is, if for two asset levels $a_t^2 \geq a_t^1$ and two lump-sum transfers $\zeta^2 \geq \zeta^1$, if two asset choices satisfy a_{t+1} and $\tilde{a}_{t+1}$ satisfy

$$a_{t+1} \leq (1 + r)a_t^1 + e_t + \zeta^2; \quad \tilde{a}_{t+1} \leq (1 + r)a_t^2 + e_t + \zeta^1 \quad (\text{A.3})$$

then it holds that

$$\min(a_{t+1}, \tilde{a}_{t+1}) \leq (1 + r)a_t^1 + e_t + \zeta^1 \quad (\text{A.4})$$

$$\max(a_{t+1}, \tilde{a}_{t+1}) \leq (1 + r)a_t^2 + e_t + \zeta^2, \quad (\text{A.5})$$

which is obviously correct. Furthermore, recognizing that u has strictly increasing differences,

$$\frac{\partial^2 u((1 + r)a_t + e_t + \zeta - a_{t+1})}{\partial a_{t+1} \partial \zeta} > 0 \quad (\text{A.6})$$

implies that an increase in ζ is a positive shock.

This result ensures that each household's asset demand is nondecreasing if lump-sum transfers are increased or equivalently is nonincreasing if lump-sum taxes are increased. If the absolute risk aversion is not constant that is either decreasing or increasing, aggregate steady-state

asset demand strictly increases. Since asset demand is nondecreasing for all households it is sufficient to show the strict increase for one individual. Consider the first-order condition of household i with the highest consumption level $c_{i,t}$ in Period t , implying that this household is unconstrained and that the consumption Euler equation holds with equality:

$$u'(c_{i,t}) = \beta(1 + r_{ss})E_{i,t}u'(c_{i,t+1}).$$

If savings were unchanged for all households, then a necessary condition for unchanged savings to be optimal for household i is

$$u''(c_{i,t}) = \beta(1 + r_{ss})E_{i,t}u''(c_{i,t+1}),$$

the derivative of the consumption Euler equation with respect to a permanent transfer increase (which is consumed in each period).

I now use that the coefficient of absolute risk aversion, $\Xi(c) = -u''(c)/u'(c)$, is not constant and either increasing or decreasing, that $c_{i,t}/c_{i,t+1} \geq 1$ with strict inequality for at least one $t + 1$ state (follows from $\beta(1 + r_{ss}) < 1$) and that

$$\begin{aligned} u'(c_{i,t}) &= \frac{-u''(c_{i,t})}{\Xi(c_{i,t})} \\ &= \beta(1 + r_{ss})E_{i,t} \frac{-u''(c_{i,t+1})}{\Xi(c_{i,t+1})} = \beta(1 + r_{ss})E_{i,t}u'(c_{i,t+1}) \frac{\Xi(c_{i,t+1})}{\Xi(c_{i,t})}. \end{aligned}$$

If $\Xi(c)$ is strictly decreasing then $\Xi(c_{i,t+1})/\Xi(c_{i,t}) \geq 1$ with strict inequality for at least one $t + 1$ state and if $\Xi(c)$ is strictly increasing then $\Xi(c_{i,t+1})/\Xi(c_{i,t}) \leq 1$ with strict inequality for at least one $t + 1$ state. Therefore,

$$u'(c_{i,t}) \begin{cases} > \beta(1 + r_{ss})E_{i,t}u'(c_{i,t+1}) = u'(c_{i,t}), \text{ if } \Xi(c) \text{ is strictly decreasing,} \\ < \beta(1 + r_{ss})E_{i,t}u'(c_{i,t+1}) = u'(c_{i,t}), \text{ if } \Xi(c) \text{ is strictly increasing,} \end{cases}$$

in both cases a contradiction, establishing that savings strictly increase for this individual and thus the partial effect of a higher lump-sum transfer(tax) on aggregate savings and asset demand is strictly positive (negative).

As emphasized in Acemoglu and Jensen (2015), the partial result does not necessarily carry over to general equilibrium. Here, the argument is, however, not difficult since the real interest rate is determined by monetary and fiscal policy and is thus fixed. Therefore, the result on aggregate asset demand now follows using the results on individual behavior. For aggregate asset demand and $P > P$ and thus $G/\bar{P} < G/P$ it follows that

$$\begin{aligned} S(1 + r_{ss}, G/\bar{P}) &= \tilde{S} = S(1 + r_{ss}, G/\bar{P} - \omega + r_{ss}\tilde{S}) > S_{ss} \\ &= S(1 + r_{ss}, G/P - \omega + r_{ss}S_{ss}) = S(1 + r_{ss}, G/P). \end{aligned}$$

To see this, suppose instead that $\tilde{S} \leq S_{ss}$, implying

$$\begin{aligned} S_{ss} &= S(1 + r_{ss}, G/P - \omega + r_{ss}S_{ss}) \\ &< S(1 + r_{ss}, G/\bar{P} - \omega + r_{ss}S_{ss}) \\ &\leq S(1 + r_{ss}, G/\bar{P} - \omega + r_{ss}\tilde{S}) \\ &= \tilde{S}, \end{aligned}$$

where the first equality is the definition of S_{ss} being the fixed point at price level P , the first inequality is the partial result of lump-sum taxes on asset demand since taxes are lower for $\bar{P} > P$ and the last inequality is the same partial result combined with the assumption that $\tilde{S} \leq S_{ss}$. This results in a contradiction, $\tilde{S} < \tilde{S}$, establishing that the fixed point $S(1 + r_{ss}, G/P) = S(1 + r_{ss}, G/P - \omega + r_{ss}S(1 + r_{ss}, G/P))$ is strictly increasing in P or equivalently strictly decreasing in G/P .

A.1.2 | Price-Level Determinacy With Natural Debt Limits (NDL)

In the main text, debt limits are exogenous. With such debt limits, the intuition (already in Bewley 1980) for the determinacy result is that with incomplete markets, the household sector effectively values real bonds, and fiscal policy determines the level of nominal government debt.

The NDL assumption differs from exogenous credit constraints, since it merely requires that the household can definitely repay the loan. In a steady state for an interest rate $r > 0$ and $e_1 = \min\{e \mid e \in \mathcal{E}\}$, this yields

$$a_{t+1} \geq -\frac{e_1}{r}. \quad (\text{A.7})$$

If the real interest rate $r < 0$, which is possible in incomplete markets models, an exogenous constraint as (2) needs to be imposed, and we are back to the exogenous constraint case. I therefore focus on the $r > 0$ case.

The NDL can give rise to a Modigliani–Miller theorem of government finance, where there is a continuum of government policies that leads to the same equilibrium consumption choices as in Wallace (1981), Chamley and Polemarchakis (1984), Peled (1985), and Sargent and Smith (1987). An incomplete markets model with NDL allows for such a theorem, where the level of real government debt is irrelevant and thus does not affect household consumption choices (Bhandari et al. 2017). This type of real indeterminacy translates into steady-state price-level indeterminacy, if there are two stationary equilibria with the same real interest rate $1 + r_{ss}$ and the same household consumption choices but with different price levels P_1 and P_2 .

To characterize the type of fiscal policies which give rise to such an indeterminacy, we define the nominal taxes of an agent with endowment e as $T(e)$. Without government expenditure, taxes are only raised to pay for interest payments on bonds, so that the steady-state government budget constraint is satisfied

$$\int T(e)d\Omega = iB. \quad (\text{A.8})$$

Consider a household in the equilibrium with price level P_1 , endowment e , and beginning-of-period real assets $a_1(e)$. Let $a_2(e)$ be real assets of this household in the other price P_2 equilibrium. For the consumption choice of this household to be the same across the two equilibria, the budget constraint has to be the same, implying³⁵

$$r_{ss}(a_1(e) - a_2(e)) = \frac{T(e)}{P_1} - \frac{T(e)}{P_2}. \quad (\text{A.9})$$

In the model with exogenous credit constraints, this condition would not leave household consumption choices unchanged. For example, consider a credit-constrained household in the P_1 equilibrium. Then, $a_2(e) < a_1(e)$ is not feasible, as it violates the credit constraint and if $a_2(e) > a_1(e)$, some of the additional wealth is consumed. With NDL, this argument could break down, as some Ricardian equivalence-type argument kicks in. With NDL, the credit constraint depends on future after-tax income and thus the credit constraint is relaxed if future tax obligations fall. If tax obligations fall enough, this could render $a_2(e) < a_1(e)$ feasible. I will show and verify below what “falling enough” means, but I proceed now just using (84) and assume that the consumption choice is unaffected. This implies that the difference in assets across the two equilibria carried to the next period is $a_1(e) - a_2(e)$. For the consumption choice in the next period to be unaffected requires again that the budget constraints be the same, implying

$$r_{ss}(a_1(e) - a_2(e)) = \frac{T(e')}{P_1} - \frac{T(e')}{P_2} \quad (\text{A.10})$$

with the important difference that taxes depend on next period's endowment e' . Iterating this argument shows that

$$r_{ss}(a_1(e) - a_2(e)) = \frac{T(\hat{e})}{P_1} - \frac{T(\hat{e})}{P_2} \quad (\text{A.11})$$

for all $\hat{e} \in \mathcal{E}$, that is the right-hand side is constant. Therefore,

$$\forall \hat{e} \in \mathcal{E} : \frac{T(e) - T(\hat{e})}{P_1} = \frac{T(e) - T(\hat{e})}{P_2}, \quad (\text{A.12})$$

implying, since $P_1 \neq P_2$, that taxes do not depend on e and are lump-sum,

$$T(e) = T. \quad (\text{A.13})$$

In line with the arguments in the literature cited above, lump-sum taxation implies that constructing a continuum of equilibria is straightforward, but more subtle than merely multiplying all prices by some number $\lambda > 0$. Start with one price level P and an associated stationary asset distribution described through Ω . We now consider a different price $\hat{P} \neq P$, and construct a different stationary equilibrium.

Define $\delta = (T/\hat{P} - T/P)/r$ and shift the distribution of assets by δ , that is, if a household holds a in the P stationary equilibrium, then it holds $\hat{a} = a + \delta$ in the new $\hat{P}$ stationary equilibrium. Household budget constraints change to

$$\begin{aligned} (1+r)\hat{a} - c + e - \frac{T}{\hat{P}} &= \\ &= (1+r)(a + \delta) - c + e - \frac{T}{\hat{P}} - r\delta = (1+r)a - c + e + \delta = a' + \delta = \hat{a}'. \end{aligned} \quad (\text{A.14})$$

That is, for the same consumption choice c , the household now carries δ more assets, $a' + \delta$, to the next period, consistent with the idea that the asset distribution is shifted for each household in all states by δ . Following Bhandari et al. (2017), it is now straightforward to show that the same consumption choices are optimal in both equilibria. The main idea is the same as in Ricardian equivalence proofs. The difference in lump-sum taxes is equal to the difference in interest payments for each household, such that a higher level of assets is merely used to cover higher tax payments without affecting consumption choices. The government budget constraint is also satisfied

$$\frac{T}{\hat{P}} = \frac{T}{P} + r\delta = r\left(\frac{B}{P} + \delta\right) = r \int (a_i + \delta) di = r \int \hat{a}_i di. \quad (\text{A.15})$$

Note that in none of these equilibria is the real value of bonds owned by a household equal to the present value of its tax obligations. The reason is simple: households hold different amounts of bonds, but all pay the same amount of taxes. Instead, the difference in the real value of a household's bonds across equilibria is equal to the difference in its present value of taxes. This also explains why it is not sufficient simply to multiply the price P with λ . Such a multiplication would change the tax obligation for everyone by the same amount, x , but the real value of household bonds is not changed by the same absolute amount, but only by the same percentage, $1/\lambda$. MPC heterogeneity implies that this redistribution of wealth does not leave consumption choices unaffected.

These arguments show that the only scenario in which the price level is indeterminate, is with NDL and lump-sum taxes as the only tax instrument. In this case, constructing aggregate steady-state asset demand as in the main text as a fixed point of

$$S = S(1+r, \tau = rS) \quad (\text{A.16})$$

is not well-defined. If S is a fixed point, then $S + \delta$ is a fixed point as well, as I have just shown.

In all other cases—exogenous borrowing constraints or non-lump-sum taxes—the price level is again determinate and the fixed point is well-defined, since no Modigliani–Miller theorem of government finance holds. Shifts in taxes affect consumption choices and thus savings decisions in such a way that no other fixed point is obtained. This latter scenario is the empirically relevant one. The NDL allocation is too close to the complete markets outcome and thus cannot match the empirical facts that motivate the heterogeneous agents model. Income heterogeneity also restricts the size of lump-sum taxes to be less than the lowest income level,

requiring a richer tax code to cover government expenditures (Werning 2007). This renders the indeterminacy case an interesting but purely theoretical possibility.

The NDL also does not lead to indeterminacy when government expenditure is (partially) nominal. In particular, the price level is determinate even with NDL, and for all tax policies, as long as government expenditure is (partially) nominal. If bonds B are nominal, taxes now satisfy

$$\frac{T}{P} = \frac{G}{P} - \omega + i\frac{B}{P}, \quad (\text{A.17})$$

such that steady-state asset demand is a fixed point of

$$S = S(1+r, \frac{G}{P} - \omega + rS). \quad (\text{A.18})$$

Although this is again a fixed-point problem, the above arguments for showing price-level indeterminacy with NDL do not apply here. To observe this, suppose to the contrary that there are two different equilibrium prices P_1 and P_2 . The same arguments as above show that a necessary condition is again that taxes are lump-sum, so that $\omega = 0$. Shifting the assets of each household by δ , where

$$\frac{B}{P_2} = \frac{B}{P_1} + \delta, \quad (\text{A.19})$$

then requires

$$\frac{T}{P_2} = \frac{T}{P_1} + \delta r \quad (\text{A.20})$$

to keep household budget constraints unaffected. But then, the steady-state government budget constraint for P_2 is not satisfied, if it is satisfied for P_1

$$\frac{G}{P_2} + r\frac{B}{P_2} - \frac{T}{P_2} \quad (\text{A.21})$$

$$= \frac{G}{P_2} + r\left(\frac{B}{P_1} + \delta\right) - \left(\frac{T}{P_1} + \delta r\right) \quad (\text{A.22})$$

$$= \frac{G}{P_2} + r\frac{B}{P_1} - \frac{T}{P_1} \quad (\text{A.23})$$

$$= \frac{G}{P_2} - \underbrace{\frac{G}{P_1} + \frac{G}{P_1}}_{=0} + r\frac{B}{P_1} - \frac{T}{P_1} \quad (\text{A.24})$$

$$= \frac{G}{P_2} - \frac{G}{P_1} + \underbrace{\frac{G}{P_1} + r\frac{B}{P_1} - \frac{T}{P_1}}_{=0} \quad (\text{A.25})$$

$$= \frac{G}{P_2} - \frac{G}{P_1} \neq 0. \quad (\text{A.26})$$

The intuition is straightforward. The shift in prices not only shifts taxes one-for-one with interest rate payments on debt, but also shifts real government spending if it is partially nominal. This leaves the household budget constraint unaffected, but not the government budget constraint, as government expenditure shows up in the latter, but not in the former constraint.

A.1.3 | Price-Level Indeterminacy: Perpetual Youth Model

Similar arguments for showing indeterminacy in complete markets models apply in “perpetual youth” models (Yaari 1965; Blanchard 1985) since the steady-state interest rate is again equal to the discount rate, but now adjusted for the probability of death or retirement, so that $(1 + r_{ss})\tilde{\beta} = 1$ in a steady state for the adjusted discount rate $\tilde{\beta}$. Again, the steady-state real interest rate is independent of the price level and only the change in prices, $1 + \pi_{ss}$, but not the level itself is determined.

In this class of models, however, this is not the only equilibrium, if the Samuelson dynamic inefficiency condition is satisfied. In this case, both a bubbleless as well as a continuum of bubbly equilibria exist, a scenario

explored in recent work by Galí (2021). Whereas most papers assume that the bubble is a real asset affecting the stock market or housing market, a monetary bubble may coexist, so that money has value as in Samuelson's work. As a result, there is a continuum of equilibria, each associated with a different value of money (= different size of monetary bubble) and each associated with a different price level. As an example, suppose that a bubble exists that has a real value of one. In one equilibrium, nominal money has a value of one, the price level is one, and there are thus no real bubbles. In another equilibrium, the price level is two and there is a real bubble with a value one half. Alternatively, the price level is three and the real bubble has a value of two thirds. Or the real bubble has a value of one and money has no value.

Bénassy (2005;2008) makes a particular choice on the size of the monetary bubble through ruling out real bubbles (the first case in the previous example) and conditional on this choice, find a unique bubbly price level. This approach however, does not overcome the indeterminacy problem in the "bubbleless" equilibrium and it rules out other bubbly equilibria with different price levels by assumption.³⁶ Bénassy (2005;2008) needs to make this equilibrium selection in order to obtain a well-defined demand for money (or more generally for nominal government liabilities), since the Samuelson logic only delivers the existence of a monetary equilibrium, but not uniqueness. This shows again, as in the Hand-to-Mouth economy, that the failure of Ricardian equivalence is a necessary but not sufficient condition for price-level determinacy.

A.1.4 | Price-Level Indeterminacy: Complete Markets and Aggregate Risk

Price-level indeterminacy also arises in representative agent economies with aggregate risk. Assume there are n aggregate shocks $s_1, \dots, s_n$ with associated consumption levels of the representative household $c_1, \dots, c_n$ and marginal utilities of consumption $u_1, \dots, u_n$. The FOCs for nominal bonds are therefore

$$\frac{u_i}{\bar{P}_i} = \beta \frac{1 + i_{ss}}{1 + \pi} \sum_{j=1}^n q_{ij} \frac{u_j}{\bar{P}_j},$$

where $q_{ij} = \text{Prob}(s_j | s_i)$, $\bar{P}_i$ is the price level in state s_i (normalized by the price trend), and the inflation rate π , which is equal to the constant growth rate of nominal debt. Since consumption c_i is equal to endowment in this state s_i , marginal utilities u_i do not depend on prices. Therefore, for each $\kappa > 0$ multiplying all prices in all states by κ is also an equilibrium, establishing indeterminacy. Adding aggregate risk to an economy with permanent income hypothesis (PIH) households and hand-to-mouth households also does not overcome the indeterminacy problem. The same arguments for the representative agent now apply to the PIH households.

Appendix B: Monetary and Fiscal Policy and Ljungqvist and Sargent's (2012) Ten Monetary Doctrines

In this section, I explain the differences between the FTPL and my theory in more detail. The starting point is Chapter 26 on "Fiscal-Monetary Theories of Inflation" in Ljungqvist and Sargent (2012), which considers a policy designed to differentiate between the initial Period $t = 0$ ("short run") and the remaining Period $t \geq 1$ ("long run"). I adopt this setting in my model and thus also allow for real government expenditures g_t , such that the Period t government budget constraint is

$$T_t := (1 + i_t)B_t + P_t g_t - B_{t+1}. \quad (\text{A.27})$$

The objective of this section is illustrating how the initial price P_0 is determined and why the two theories, FTPL and DTPL, deliver different P_0 .

I first consider the DTPL and assume that

$$g_t = g \quad \forall t \geq 0 \quad (\text{A.28})$$

$$B_t = B \quad \forall t \geq 1$$

$$T_t = T \quad \forall t \geq 1,$$

$$i_t = i \quad \forall t \geq 0,$$

where I permit initial bonds $B_0 \neq B$ and therefore also initial lump-sum taxes $T_0 \neq T$. The steady-state inflation rate is $\pi_{ss} = 0$. I use the asset demand function $S_{t+1}(\Omega_t, (1+i)/P_t, 1+r_{t+1}, 1+r_{t+2}, \dots; \tau_t, \tau_{t+1}, \dots)$ but with some modifications to simplify solving the model backward. Ω_t is now the joint distribution of nominal assets and productivity at the beginning of Period t and asset demand S_{t+1} depends on $(1+i)/P_t$. Asset demand S_{t+1} still depends on the sequence of future real interest rates and taxes.³⁷

Using $1 + r_{t+k+1} = (1+i)P_{t+k}/P_{t+k+1} \forall k \geq 0$ and the government budget constraint (102), the equilibrium asset-market clearing condition in Period t is then

$$S_{t+1}(\Omega_t, \frac{1+i}{P_t}, (1+i)\frac{P_t}{P_{t+1}}, (1+i)\frac{P_{t+1}}{P_{t+2}}, \dots; \quad (\text{A.29})$$

$$g_t + \frac{(1+i)B_t - B_{t+1}}{P_t}, g_{t+1} + \frac{(1+i)B_{t+1} - B_{t+2}}{P_{t+1}}, \dots) - \frac{B_{t+1}}{P_t} = 0.$$

Assume now that such an equilibrium sequence of asset demands, distributions, and prices exists and that a steady state is reached after N periods. Given this sequence of equilibrium distributions and invoking some further assumptions to illustrate the mechanism, the sequence of prices including the initial price level P_0 can be determined backward.³⁸ For the sake of argument, assume that condition (104) is upward sloping in the price P_t (Hagedorn (2023) provides conditions for this to be true), implying a unique solution P_t , taking all other future prices as given. Starting with the steady-state price level $P_N = P^*$, this argument allows computing the full sequence of prices backward. First, the price P_{N-1} is calculated, taking P_N and the equilibrium distribution Ω_{N-1} as given as the solution to

$$S_{N-1}(\Omega_{N-1}, \frac{1+i}{P_{N-1}}, (1+i)\frac{P_{N-1}}{P_N}, 1+r_{ss}, \dots; g_t + i\frac{B}{P_{N-1}}, g + i\frac{B}{P_N}, \dots) - \frac{B}{P_{N-1}} = 0.$$

In the next step, P_{N-2} is the solution, given the equilibrium distribution Ω_{N-2} , to

$$S_{N-1}(\Omega_{N-2}, \frac{1+i}{P_{N-2}}, (1+i)\frac{P_{N-2}}{P_{N-1}}, (1+i)\frac{P_{N-1}}{P_N}, 1+r_{ss}, \dots; g_t + i\frac{B}{P_{N-2}}, g + i\frac{B}{P_{N-1}}, \dots) - \frac{B}{P_{N-2}} = 0.$$

This procedure is iterated until the initial price level P_0 is computed. If $B_0 = B$, then

$$P_t = P^* \quad \forall t \geq 0. \quad (\text{A.30})$$

The Fiscal Theory of the Price Level (FTPL) takes a different approach. To clarify the difference, now I assume that markets are complete and that lump-sum taxes are in real terms and fixed at τ , whereas the amount of government bonds B_t , $t \geq 1$ is endogenous. The initial debt level B_0 is again inherited from the past and $g_t = g$, $i_t = i$, $\forall t \geq 0$. The Period t government budget constraint in real term reads

$$\frac{B_{t+1}}{P_t} = (1 + r_t) \frac{B_t}{P_{t-1}} + g - \tau. \quad (\text{A.31})$$

Since aggregate endowment and thus consumption are constant, the real interest rate $1 + r_t = 1 + r$ is also constant. The intertemporal government budget constraint states that

$$\frac{B_1}{P_0} = \sum_{t=1}^{\infty} \frac{1}{(1+r)^t} (\tau - g) \quad (\text{A.32})$$

and thus, we obtain for the initial debt level B_0

$$\frac{B_0}{P_0} = \frac{1}{1+i} \left[\frac{B_1}{P_0} + (\tau - g) \right] = \frac{1}{1+i} \sum_{t=0}^{\infty} \frac{1}{(1+r)^t} (\tau - g) = \frac{(\tau - g)(1+r)}{r(1+i)}, \quad (\text{A.33})$$

which using the complete markets property $1 + r = 1/\beta$, yields

$$\frac{B_0}{P_0} = \frac{\tau - g}{(1+i)(1-\beta)}. \quad (\text{A.34})$$

The FTPL assumes that g and τ are exogenous, so that the initial price level P_0^{FTPL} is determined as the ratio of outstanding nominal debt B_0 to the present value of the primary surplus. Note that future debt levels B_t ($t \geq 1$) are determined from the flow government budget constraint (106).

Apparently, the determination of the price level when markets are incomplete, or when the FTPL is operating, are different. In the first case, the asset-market clearing condition is used, so that the price level depends on asset demand. In the second case, the government budget constraint is used, so that the price level depends on the initial outstanding government debt level B_0 and the fiscal variables g and τ .

If $B_0 = B$, the difference becomes even clearer. Suppose that both theories delivered the same price level, $P^* = P^{FTPL}$. Then

$$S(1+r) = \frac{B}{P^*} = \frac{B}{P^{FTPL}} = \frac{(\tau - g)(1+r)}{r(1+i)} \quad (\text{A.35})$$

implying that steady-state asset demand, using $1 + r = 1 + i$, equals

$$S(1+r) = S(1+i) = \frac{\tau - g}{i}, \quad (\text{A.36})$$

which is generically not true. For example, it is well-known that asset demand converges to infinity if $1 + r \rightarrow 1/\beta$, whereas the right side does not,

$$\lim_{1+i \rightarrow 1/\beta} S(1+i) = \infty > \lim_{1+i \rightarrow 1/\beta} \frac{\tau - g}{i}. \quad (\text{A.36})$$

The reason for the difference is that P^* is determined as clearing the asset market and that the price P_0^{FTPL} is determined as satisfying the government budget constraint. This constraint also has to be satisfied in the first case, although it is not the price, but the lump-sum tax T which ensures this, such that both the asset market clears and the budget constraint is satisfied when markets are incomplete. This means that fiscal policy is passive in the first case, whereas it is active in the FTPL (Leeper 1991).

Both constraints are also satisfied when the FTPL is operating. The government budget constraint is satisfied by construction. The asset market clears as well, but in the context of the FTPL markets being complete and asset market clearing only pinning down the real interest rate $1 + r = 1/\beta$. This is a consequence of Ricardian equivalence, since the private sector is willing to absorb any equilibrium amount of real bonds. Using the FTPL in different environments in which Ricardian equivalence does not hold, for example, if markets are incomplete, requires the private sector to be willing to absorb the real value of debt, which at the same time satisfies the government budget constraint.